

Math 6A
Discrete Math

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Chapter 1

Section 1 - Intro to Logic and Proofs

1.1 Section 1.1 - Intro to Logic

1.1.1 Propositions

Definition 1 (Proposition)

A **proposition** is a statement that is either

Example 1 (Examples with Propositions)

Determine if the following are propositions

1. It's all coming back to me now

Solution:

yes

2. Hello.

Solution:

No

3. $x + 3 = 10$

Solution:

No, since we don't know x

4. In a right triangle, $a^2 + b^2 = c^2$

Solution:

No, since a, b, c are not assigned anything

5. In a right triangle having legs of length a and b , and hypotenuse of c , then $a^2 + b^2 = c^2$

Solution:

Yes, since a, b, c are properly specified

6. There are 75 people in this room or the sky is purple

Solution:

Yes

Note

7 is a **compound proposition**

Definition 2 (Compound Proposition)

A **compound proposition** is a proposition that contains two or more **atomic** propositions. In the previous example, the two are:

1. There are 75 people in this room
2. The sky is purple

Definition 3 (Atomic Proposition)

An **atomic proposition** is a proposition that cannot be broken down further into separate propositions

Note

We denote separate propositions as: p, q, r

1.1.2 Logical Operations (connectives)

Name	Notation	Phrasing	Truth Value
Disjunction (or)	$p \vee q$	"p or q"	True when p, q or both are TRUE
Conjunction (and)	$p \wedge q$	"p and q"	True when both are TRUE
Negation (not)	$\neg p$	"not p"	The opposite truth value of p
Exclusive Or (XOR)	$p \oplus q$	"p or q, but not both"	TRUE when only one is TRUE

Here are the truth tables of each operation

p	q	$p \vee q$	$p \wedge q$	$\neg p$	$p \oplus q$
T	T	T	T	F	F
T	F	T	F	F	T
F	T	T	F	T	T
F	F	F	F	T	F

Example 2 (Logical Operators Examples)

Let p be the proposition "a is odd" and q be "a is even" where $a = 3$

p : 3 is odd = **TRUE**

q : 9 is even = **FALSE**

Write the following Propositions:

1. $\neg p$ (3 is even)

Solution:

FALSE

2. $\neg q$ (9 is odd)

Solution:

TRUE

3. $p \vee q$ (3 is odd or 9 is even)

Solution:

TRUE since 3 is odd

4. $p \wedge q$ (3 is odd and 9 is even)

Solution:

FALSE since 9 is not even

5. $p \wedge \neg q$ (3 is odd and 9 is odd)

Solution:

TRUE since both 3 is odd and 9 is odd

1.1.3 Conditionals

Name	Notation	Phrasing	Truth Value
Conditional	$p \Rightarrow q$	"p implies q" or "if p then q"	Always True except when $T \Rightarrow F$
Biconditional	$p \Leftrightarrow q$	"p if and only if q"	True when p and q are the same value

Here are the truth tables

p	q	$p \Rightarrow q$	$p \Leftrightarrow q$
T	T	T	T
T	F	F	F
F	T	T	F
F	F	T	T

1.1.4 Logical Transforms

Name	Original	Output
Converse (swap)	$p \Rightarrow q$	$q \Rightarrow p$
Inverse (negate)	$p \Rightarrow q$	$\neg p \Rightarrow \neg q$
Contrapositive (converse + inverse)	$p \Rightarrow q$	$\neg q \Rightarrow \neg p$

Here are the truth tables

p	q	$p \Rightarrow q$	$\neg q$	$\neg p$	$\neg q \Rightarrow \neg p$
TRUE	TRUE	TRUE	FALSE	FALSE	TRUE
TRUE	FALSE	FALSE	TRUE	FALSE	FALSE
FALSE	TRUE	TRUE	FALSE	TRUE	TRUE
FALSE	FALSE	TRUE	TRUE	TRUE	TRUE

Proposition 1 (Contrapositive Correlation)

From the truth table, we see that the converse is logically equivalent to the original expression, so

$$p \Rightarrow q \equiv \neg q \Rightarrow \neg p$$

Definition 4 (Logical Equivalence)

Two propositions are **logically equivalent** if they have the same truth tables

Example 3 (Logical Transforms Examples)

Let p be the proposition "it's cold outside" and q be "it's snowing outside"

1. $p \implies q$ (if it's cold outside, then it's snowing outside)

Solution:

FALSE

2. $q \implies p$ (converse) (if it's snowing outside, then it's cold outside)

Solution:

TRUE

3. $\neg p \implies \neg q$ (inverse) (if it's not cold outside, then it's not snowing)

Solution:

TRUE

4. $\neg q \implies \neg p$ (contrapositive) (if it's not snowing outside, then it's not cold outside)

Solution:

FALSE

Example 4 (More Examples with Logical Transforms)

Let p be the proposition "The triangle is equilateral" and q be "The triangle is equiangular"

1. $p \implies q$ (if the triangle is equilateral, then triangle is equiangular)

Solution:

TRUE

2. $q \implies p$ (converse) (if the triangle is equiangular, then the triangle is

equilateral)

Solution:

TRUE

3. $\neg p \implies \neg q$ (inverse) (if the triangle is not equiangular, then the triangle is not equilateral)

Solution:

TRUE

4. $\neg q \implies \neg p$ (contrapositive) (if the triangle is not equiangular then the triangle is not equilateral)

Solution:

TRUE

Example 5 (Truth Table Examples)

1. $\neg p \vee q$

p	q	$\neg p$	$\neg p \vee q$
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	T

2. $(p \implies r) \wedge (q \implies r)$

im too lazy to fill this out

1.3 Section 1.3 - Logical Equivalence

Recall 1 (Logical Equivalence)

Recall that two propositions are **logically equivalent** if they have the same truth table

1.3.1 Testing for Equivalence using Truth Tables

Example 6

Determine if the following are logically equivalent

1. $p \implies q$ and $\neg p \vee q$

Solution:

p	q	$p \implies q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

$$\therefore p \implies q \equiv \neg p \vee q$$

2. $\neg(p \wedge q)$ and $\neg p \vee \neg q$

Solution:

p	q	$p \wedge q$	$\neg(p \wedge q)$	$\neg p$	$\neg q$	$\neg p \vee \neg q$
T	T	T	F	F	F	F
T	F	F	T	F	T	T
F	T	F	T	T	F	T
F	F	F	T	T	T	T

$$\therefore \neg(p \wedge q) \equiv \neg p \vee \neg q$$

3. $(p \implies q) \vee (q \implies r)$ and $p \implies (q \vee r)$

Solution:

p	q	r	$p \implies r$	$q \implies r$	$(p \implies r) \vee (q \implies r)$	$q \vee r$	$p \implies (q \vee r)$
T	T	T	T	T	T	T	T
T	T	F	F	F	F	T	T
T	F	T	T	T	T	T	T
T	F	F	F	T	T	F	F
F	T	T	T	T	T	T	T
F	T	F	T	F	T	T	T
F	F	T	T	T	T	T	T
F	F	F	T	T	T	F	T

$$\therefore (p \implies q) \vee (q \implies r) \not\equiv p \implies (q \vee r)$$

So far we know the following:

- $p \implies q \equiv \neg p \vee q$
- DeMorgan's Law
 - $\neg(p \vee q) \equiv \neg p \wedge \neg q$
 - $\neg(p \wedge q) \equiv \neg p \vee \neg q$
- Commutative (swap places)
 - $p \vee q \equiv q \vee p$
 - $p \wedge q \equiv q \wedge p$
- Associative (swap grouping)
 - $(p \vee q) \vee r \equiv p \vee (q \vee r)$
 - $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
- Distributive (distributing operations)
 - $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
 - $p \wedge (q \vee r) \equiv (p \wedge q) \vee (p \wedge r)$
- Identity
 - $p \wedge T \equiv p$
 - $p \vee F \equiv p$
- Domination Laws
 - $p \vee T \equiv T$
 - $p \wedge F \equiv F$
- Impotent Laws
 - $p \vee p \equiv p$

$$- p \wedge p \equiv p$$

Definition 5 (Tautology and Contradiction)

A proposition that is always **TRUE** is a **tautology**

A proposition that is always **FALSE** is a **contradiction**

Note

When proving tautologies or contradictions, you could either:

1. Use Truth Tables
2. Use the Laws from earlier to reduce a compound proposition into **TRUE** or **FALSE**

But it's advised to use Truth Tables, since it's easy to get lost in your work when using the Laws

Also, ALWAYS start by **converting conditionals**

Example 7 (Examples of Tautologies and Contradictions)

Determine if the following are tautologies, contradictions, or neither

1. $(p \wedge q) \implies q$

Proving with Truth Tables

p	q	$p \wedge q$	q	$(p \wedge q) \implies q$
T	T	T	T	T
T	F	F	F	T
F	T	F	T	T
F	F	F	F	T

$\therefore (p \wedge q) \implies q \equiv \text{tautology}$

Proving with Logical Equations

$$\begin{aligned}
 (p \wedge q) \implies q &\equiv \neg(p \wedge q) \vee q && \text{(Convert Conditional)} \\
 &\equiv (\neg p \vee \neg q) \vee q && \text{(DeMorgan's Law)} \\
 &\equiv \neg p \vee (\neg q \vee q) && \text{(Associative)} \\
 &\equiv \neg p \vee T && \\
 &\equiv T && \text{(Domination)}
 \end{aligned}$$

$\therefore (p \wedge q) \implies q \equiv \text{tautology}$

2. There's another example but I'm too lazy

Note

When proving the **logical equivalence** of two propositions, start with the **MORE COMPLICATED** proposition

Example 8 (Proving Logical Equivalence of Two Propositions)

Determine if $(p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$

$(p \implies r) \wedge (q \implies r)$ seems more complicated, so we'll start with that

$$\begin{aligned}(p \implies r) \wedge (q \implies r) &\equiv (\neg p \vee r) \wedge (\neg q \vee r) && \text{(Convert Conditionals)} \\ &\equiv (\neg p \wedge \neg q) \vee r && \text{(Un-Distribute)} \\ &\equiv \neg(p \vee q) \vee r && \text{(DeMorgan's Law)} \\ &\equiv (p \vee q) \implies r && \text{(Convert back to Conditional)}\end{aligned}$$

$$\therefore (p \vee q) \implies r \equiv (p \implies r) \wedge (q \implies r)$$

1.4 Section 1.4 - Propositional Functions and Quantifiers

First, let's start with an example

Example 9

Is $x + 3 = 10$ a proposition?

Solution:

No, since x is not known

However, if x is known, then $x + 3 = 10$ is a proposition

The truth value depends on the value of x

Definition 6 (Propositional Function)

We call $x + 3 = 10$ a **propositional function** since the truth value of the proposition depends on the value of x

Example 10 (Examples of Propositional Functions)

For the given propositional functions, determine the truth value for the given value of variable(s)

1. $P(x) : x + 3 = 10$

(a) $P(6)$

Solution:

FALSE since $6 + 3 \neq 10$

(b) $P(7)$

Solution:

TRUE since $7 + 3 = 10$

2. $P(a, b, c) : a^2 + b^2 = c^2$

(a) $P(11, 60, 61)$

Solution:

TRUE

$$11^2 + 60^2 = 121 + 3600 = 3721$$

$$61^2 = (60 + 1)^2 = 3600 + 120 + 1 = 3721$$

(b) $P(147, 201, 513)$

Solution:

FALSE

$$147^2 + 201^2 = \text{odd} + \text{odd} = \text{even}$$

$$513^2 = \text{odd} \neq \text{even}$$

Example 11 (Examples of Notation) 1. Is "there exists a real number x such that $x + 3 = 10$?" a proposition?

Solution:

Yes, and this proposition is **TRUE** since $x = 7$ works

We can also rewrite this as:

$$\exists x \in \mathbb{R} \mid x + 3 = 10$$

2. Is "for all real numbers x , $x + 3 = 10$ " a proposition?

Solution:

Yes, but the proposition is **FALSE** since $x = 6$ doesn't work

We can also rewrite this as

$$\forall x \in \mathbb{R} \mid x + 3 = 10$$

Definition 7 (Quantifiers)

The **existential quantifier** $\exists$ means "there exists"

The **universal quantifier** $\forall$ means "for all" or "for every"

Example 12 (Examples of Quantifiers)

Write each proposition using quantifiers, and determine their truth values

1. There exists an integer x such that $x^2 + 2 > 3x$

Solution:

$$\exists x \in \mathbb{Z} \mid x^2 + 2 > 3x$$

TRUE since $x = 3$ gives $9 + 2 > 9$

2. For all real numbers x , $x^2 + 2 > 3x$

Solution:

$$\forall x \in \mathbb{R} \mid x^2 + 2 > 3x$$

FALSE since $x = 1$ gives $1 + 2 > 3$

Note

We only need one counter example to disprove a proposition that has $\forall$

3. There exists an integer x such that $x^3 = 31687$

Solution:

$$\exists x \in \mathbb{Z} \mid x^3 = 31687$$

FALSE since $\sqrt[3]{31687} \notin \mathbb{Z}$

Note

We needed to verify that every integer does not satisfy the proposition in order to prove it

The negations of each quantifier are as follows:

$$\neg \forall \equiv \exists$$

$$\neg \exists \equiv \forall$$

Example 13 (Negation of Quantifiers)

Give the negation of the following propositions:

1. $\forall x \in \mathbb{R} \mid x^2 + 1 > 0$ **TRUE**

Solution:

$$\exists x \in \mathbb{R} \mid x^2 + 1 \leq 0$$
 FALSE

2. $\exists a \in \mathbb{Z} \mid a^3 + 2 = 3a$ **TRUE** ($a = 1$)

Solution:

$$\forall a \in \mathbb{Z} \mid a^3 + 2 \neq 3a$$
 FALSE

3. For every real number x , $x^2 + 30 < 11x$ **FALSE** ($x = 1$)

Solution:

$$\exists x \in \mathbb{R} \mid x^2 + 30 \geq 11x$$
 TRUE

4. For every prime integer $p > 3$, $p^2 - 1$ is divisible by 24 **TRUE**

Solution:

There exists a prime integer $p > 3$, $p^2 - 1$ is not divisible by 24 **FALSE**

Work

$$p^2 - 1$$

We want $p^2 - 1$ divisible by both 8 and 3

$$(p^2 - 1) = (p - 1)(p + 1)$$

$$(p - 1)(p + 1) = (\text{even})(\text{even})$$

- p is prime
- $p > 3$
- p is odd

Example 14

Statement	When TRUE	When FALSE
$\forall x, P(x)$	$P(x)$ is TRUE for all x	There exists at least one x such that $P(x)$ is FALSE
$\exists x, P(x)$	There exists at least one x such that $P(x)$ is TRUE	For all x such that $P(x)$ is FALSE

Example 15 (Examples with Negation)

Statement	Equivalent	When TRUE	When FALSE
$\neg \forall x, P(x)$	$\exists x, \neg P(x)$	There is at least one x such that $P(x)$ is FALSE	For all x , $P(x)$ is TRUE
$\neg \exists x, P(x)$	$\forall x, \neg P(x)$	$P(x)$ is FALSE for all x	There exists at least one x such that $P(x)$ is TRUE

Example 16 (Examples of Rewriting with Quantifiers)

Rewrite the following with Quantifiers

1. For all positive integers, either $6n + 1$ or $6n - 1$ is prime

Solution:

$\forall n \in \mathbb{Z}^+, 6n - 1$ is prime $\vee 6n + 1$ is prime

FALSE since if $n = 20$ then $6n - 1 = 119$ and $6n + 1 = 121$ are **composite** (not prime)

2. There exists an integer n such that $n^2 + n + 41$ is composite

Solution:

$\exists n \in \mathbb{Z}, n^2 + n + 41$ is composite

TRUE since if $n = 41$ then

$$\begin{aligned} n^2 + n + 41 &= 41^2 + 41 + 41 \\ &= 41(41 + 1 + 1) \end{aligned}$$

which is composite

1.5 Section 1.5 - Nested Quantifiers

Let $\mathbb{P}$ be the set of prime integers

$$\mathbb{P} = \{2, 3, 5, 7, 11, \dots\}$$

With this in mind, we can write the following in two ways:

Consider the statement "for all positive integers n , $6n - 1$ or $6n + 1$ is prime"

1. $\forall n \in \mathbb{Z}^+, 6n - 1 \in \mathbb{P} \vee 6n + 1 \in \mathbb{P}$
2. $\forall n \in \mathbb{Z}^+ \exists p \in \mathbb{P}, 6n - 1 = p \vee 6n + 1 = p$

Example 17 (Example of Nested Quantifiers) 1. For all real x there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R} \exists y \in \mathbb{R}, xy = 1$$

FALSE, since if $x = 0$, then $xy = 0 \neq 1$

2. For all positive real x , there exists a real number y such that $xy = 1$

$$\forall x \in \mathbb{R}^+ \exists y \in \mathbb{R}, xy = 1$$

TRUE since if $x \in \mathbb{R}^+$, then $x > 0$, and if $y = \frac{1}{x}$, then $xy = x \frac{1}{x} = 1$

Note

Given x , we can find y

y depends on x

But what happens if we switch $\forall$ and $\exists$?

3. $\exists y \in \mathbb{R} \forall x \in \mathbb{R}^+, xy = 1$

FALSE since if $x = 1$, then $y = 1$, but if $x = 2$, then $y = \frac{1}{2}$, and $1 \neq \frac{1}{2}$

Note

There is a single y that works **FOR ALL** x

y does not depend on x

4. For all positive integers a and b , we have $ab > 0$

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, ab > 0$$

Note

This is the same as $\forall a, b \in \mathbb{Z}^+$

TRUE since $a, b \in \mathbb{Z}^+$, then $a > 0$ and $b > 0$, hence $ab > 0$

5. There exists integers a , and b such that $a^2 + b^2 = 464$

$$\exists a \in \mathbb{Z} \exists b \in \mathbb{Z}, a^2 + b^2 = 464$$

$$\exists b \in \mathbb{Z} \exists a \in \mathbb{Z}, a^2 + b^2 = 464$$

TRUE $a = 8, b = 20,$

$$\begin{aligned} a^2 + b^2 &= 8^2 + 20^2 \\ &= 64 + 400 \\ &= 464 \end{aligned}$$

Note

The order of the **SAME** quantifier doesn't matter, only for **DIFFERENT** quantifiers does it matter

Example 18

Rewrite the following using Quantifiers

1. "There exists positive integers a and b such that $a^2 + b^2 = 15625$ "

(a) Write as Quantifier

$$\exists a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+, a^2 + b^2 = 15625$$

(b) Write the negation

$$\forall a \in \mathbb{Z}^+ \forall b \in \mathbb{Z}^+, a^2 + b^2 \neq 15625$$

Work

$$a^2 + b^2 = 5^6$$

$$a^2 + b^2 = 5^4 \cdot 5^2$$

$$3^2 + 4^2 = 5^2$$

$$5^4(3^2 + 4^2) = 5^4 \cdot 5^2$$

$$5^4 \cdot 3^2 + 5^4 \cdot 4^2 = 5^6$$

$$(5^2 \cdot 3)^2 + (5^2 \cdot 4)^2 = 5^6$$

$$(75)^2 + (100)^2 = 5^6$$

$$\therefore a = 75, b = 100$$

TRUE

There's more examples but I'm just too lazy to write them if you get the chance, they're on page 3 of your notes

1.7 Section 1.7 Intro to Proofs

What do we prove? Usually that some pattern or formula is true.

Example 19

$$\begin{aligned}
6 + 8 &= 14 \\
12 + 16 &= 28 \\
2 + 2 &= 4
\end{aligned}$$

We believe that the sum of 2 even integers is even.

Try to prove that: even + even = even

Work An even number can be defined as $2n$, where n is an integer

$$2n + 2m = 2(n + m)$$

Proof. let a, b be even integers.

Then there exists integers n, m such that $a = 2n, b = 2m$

$$a + b = 2n + 2m = 2(m + n)$$

is even, as desired

□

But writing it this way is too long. Let's define some notation:

1.7.1 Set Notation

Name	Symbol	Set Notation
Set of all natural numbers	$\mathbb{N}$	$\{0, 1, 2, 3, \dots\}$
Set of all integers	$\mathbb{Z}$	$\{0, -1, 1, -2, 2, \dots\}$
Set of positive integers	$\mathbb{Z}^+$	$\{1, 2, 3, \dots\}$
Set of negative integers	$\mathbb{Z}^-$	$\{-1, -2, -3, \dots\}$
Set of rational numbers	$\mathbb{Q}$	$\{\frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0\}$
Set of real numbers	$\mathbb{R}$	Limit of all convergent sequences of real numbers
Set of all complex numbers	$\mathbb{C}$	$\{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$

$\exists$ means there exists

Let's rewrite the previous example using this new notation

$a, b \in \mathbb{Z}$, both even

Then, $\exists m, n \in \mathbb{Z}$ such that

$$a = 2m, b = 2n$$

Their sum $a + b = 2m + 2n = 2(m + n)$
is even, as desired

1.7.2 Methods of Proving $p \implies q$

1. **Direct Proof:**

Start with p and *math* then conclude q

2. **Indirect Proof:**

Prove the contrapositive proof $\neg q \implies \neg p$

3. **Finite Cases:**

Use the fact that $p \equiv p_1 \vee p_2 \vee \dots \vee p_n$

Then prove $(p_1 \implies q) \wedge (p_2 \implies q) \wedge \dots \wedge (p_n \implies q)$

Chapter 2

Section 2 - Structures: Sets, Functions, Sequences, Sums, and Matrices

2.1 Section 2.1 - The Joy of Sets

Definition 8 (Sets and Elements)

A **set** is an unordered collection of objects called **elements**

$$\mathbb{N} = \{1, 2, 3, 4, \dots\} \quad (\text{Natural Numbers})$$

$$\text{Whole Numbers} = \{0, 1, 2, 3, \dots\}$$

$$\mathbb{Z} = \{0, -1, 1, -2, 2, -3, 3, \dots\}$$

$$\mathbb{Q} = \left\{ \frac{a}{b} \mid a, b \in \mathbb{Z}, b \neq 0 \right\}$$

$$\mathbb{R} = \text{limit of all convergent sequences of rational numbers}$$

$$\mathbb{C} = \{a + bi \mid a, b \in \mathbb{R}, i^2 = -1\}$$

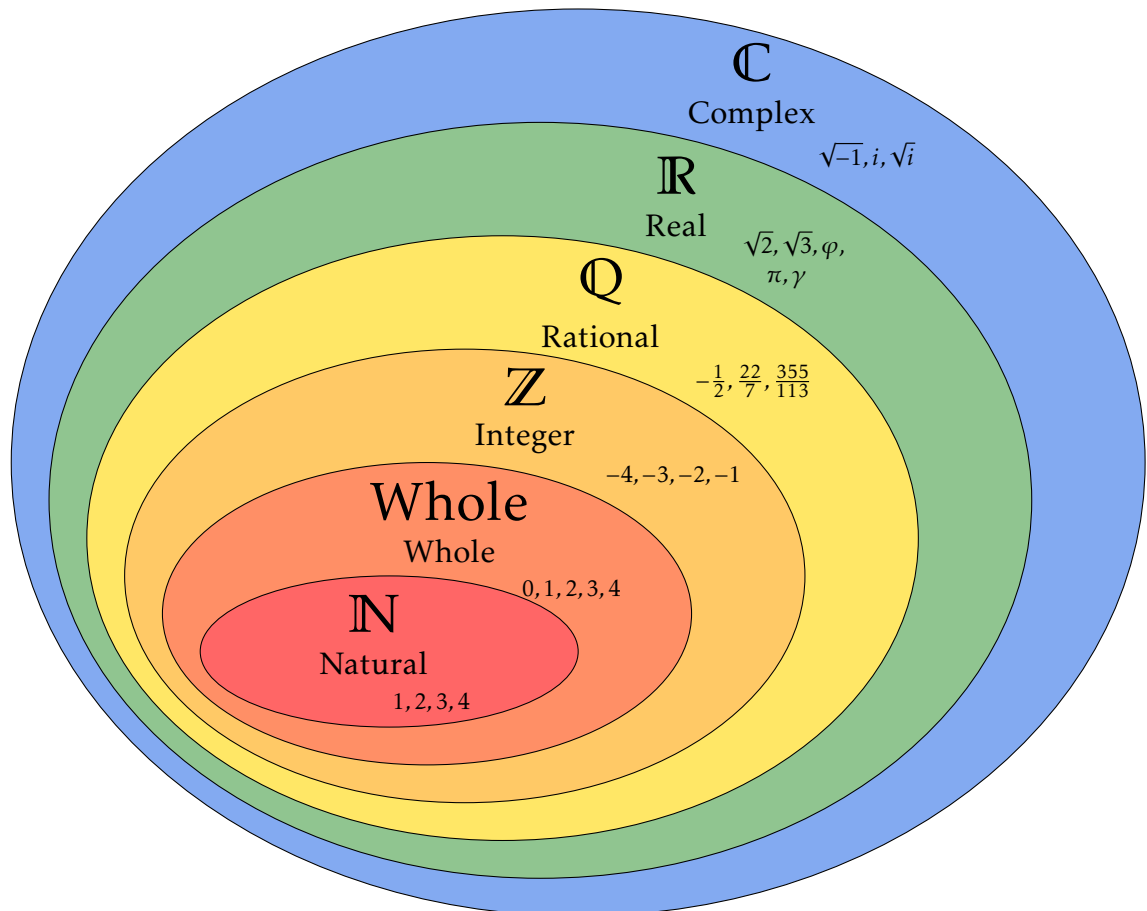
Note

This notation for $\mathbb{Q}$ is called **Set Builder Notation**

It's read as "Q is the set $\frac{a}{b}$ such that $a, b \in \mathbb{Z}, b \neq 0$ "

Note

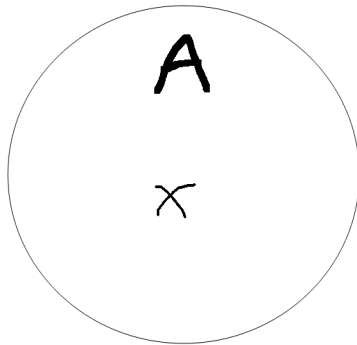
Notice that every natural number is a whole number. So we say "the set of natural numbers is a **subset** of the set of whole numbers"



Definition 9 (Elements in Sets)

If x is an element of the set A , then we write

$$x \in A$$



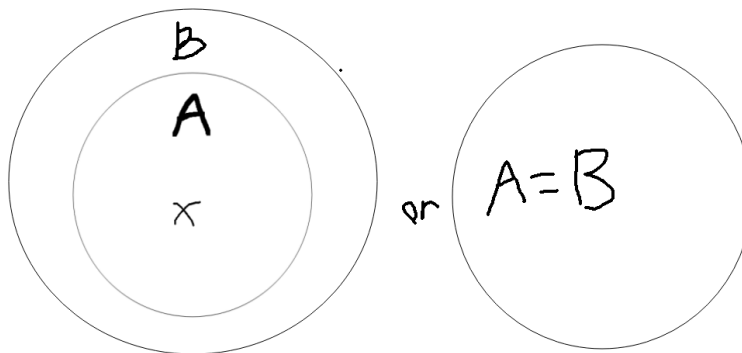
Definition 10 (Empty Set)

The set containing no elements is called the **empty set**, and is denoted by $\{\}$ or $\emptyset$

Given sets A and B such that for all $x \in A$ we have $x \in B$, then we say **A is a subset of B** and we write

$$A \subseteq B$$

Which can mean two things:



Definition 11 (Proper Sets)

If there exists $x \in B$ with $x \notin A$, then we say that A is a **proper subset** of B and we write

$$A \subsetneq B$$

Definition 12 (Cardinality)

Given a set A , the **cardinality** of a set is the number of elements in the set A , denoted by

$$|A|$$

Definition 13 (Power Set)

The **power set** of a given set S is the set of all subsets of S , denoted by

$$\mathcal{P}(S)$$

$$|\mathcal{P}(S)| = 2^{|S|}$$

Note

This includes the empty set, $\{\}$

Definition 14 (Cartesian Product)

Given sets A, B , the **cartesian product** is the set

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

$$|A \times B| = |A| \cdot |B|$$

$$A^2 = A \times A$$

Example 20

Let $A = \{1, 2, 3\}, B = \{1, 2\}$

(a) Is $A \subseteq B$?

Solution:

No since $3 \in A, 3 \notin B$

(b) Is $B \in A$?

Solution:

Yes since for all $x \in B, x \in A$

(c) What is the power set of B ?

Solution:

$$\mathcal{P} = \{\{\}, \{1\}, \{2\}, \{1, 2\}\}$$

(d) What is the power set of A ?

Solution:

$$\mathcal{P}(A) = \{\{\}, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\}\}$$

Note

If A is a finite set (ie $|A|$ is finite), then $|\mathcal{P}(A)| = 2^{|A|}$

(e) Determine $A \times B$

Solution:

$$A \times B = \{(1, 1), (1, 2), (2, 1), (2, 2), (3, 1), (3, 2)\}$$

$$B \times A = \{(1, 1), (1, 2), (1, 3), (2, 1), (2, 2), (2, 3)\}$$

$$A \times B \neq B \times A$$

$$|A \times B| = |B \times A| = |A| \cdot |B|$$

2.2 Section 2.2 - Set Operations

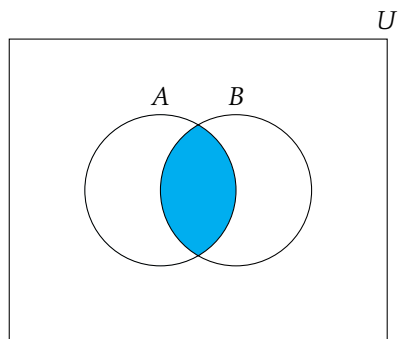
Let's consider a venn diagram which is a way of visualizing sets

Let U represent the universal set

Definition 15 (Intersection)

Elements that are in **both** A and B is called the intersection of A and B

$$A \cap B$$



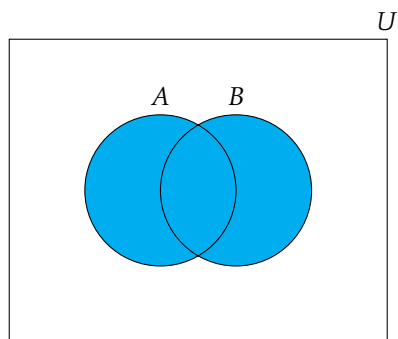
Definition 16 (Union)

Elements that are in **either** A or B is called the **union** of A and B

$$A \cup B$$

Also

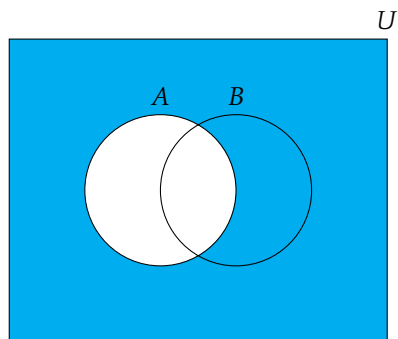
$$|A \cup B| = |A| + |B| - |A \cap B|$$



Definition 17 (Compliment)

Elements that just aren't in A is called the **compliment** of A

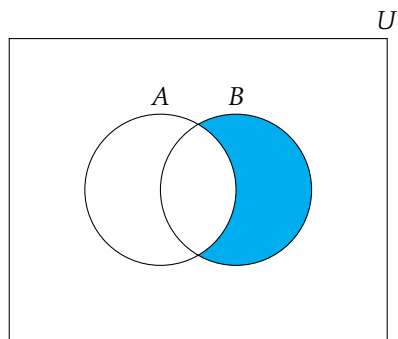
$$\bar{A}$$



Definition 18 (Difference)

Elements in B but not in A is called the **difference**

$$B - A$$



With sets, we have 3 operations:

1. Compliment $\bar{A}$ (analogous to $\neg$)
2. Union $A \cup B$ (analogous to $\vee$)
3. Intersection $A \cap B$ (analogous to $\wedge$)

Example 21

Let

$$U = \{x \in \mathbb{Z} \mid 0 \leq x \leq 20\}$$

$$A = \{x \in U \mid x \text{ is even}\}$$

$$B = \{x \in U \mid x \text{ is divisible by 3}\}$$

Determine the following

1. $A = \{0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20\}$
2. $B = \{0, 3, 6, 9, 12, 15, 18\}$
3. $A \cap B = \{0, 6, 12, 18\}$ (what they have in common)
4. $A \cup B = \{0, 2, 3, 4, 6, 8, 9, 10, 12, 14, 15, 16, 18, 20\}$ (whats in either one)
5. $\bar{A} = \{1, 3, 5, 7, 9, 11, 13, 15, 17, 19\}$ (whats not in A)
6. $B - A = \{3, 9, 15\}$ (what's in B thats not in A)
7. $|A| = 11$ (number of elements in A)
8. $|B| = 7$ (number of elements in B)
9. $|A \cup B| = 14$
10. $|A \cap B| = 4$ (number of elements that A and B have in common)
11. $|A| + |B| - |A \cap B| = 11 + 7 - 4 = 14 = |A \cup B|$

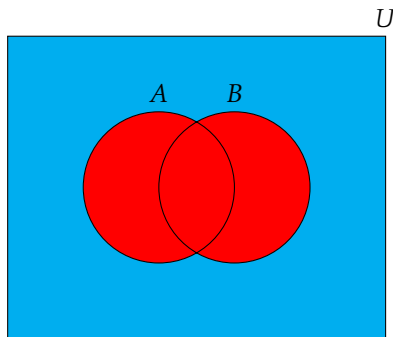
But how do these operations interact together?

Example 22

- $\overline{A \cup B}$

$$\text{Red} = A \cup B$$

$$\text{Blue} = \overline{A \cup B}$$

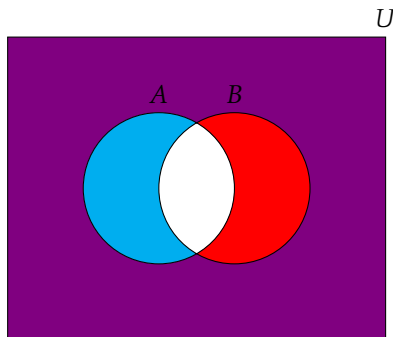


- $\bar{A} \cap \bar{B}$

$$\text{Red} = \bar{A}$$

$$\text{Blue} = \bar{B}$$

$$\text{Purple} = \bar{A} \cap \bar{B}$$



$$\therefore \overline{A \cup B} = \overline{A} \cap \overline{B}$$

1. **DeMorgan's Laws**

- $\overline{A \cup B} = \overline{A} \cap \overline{B}$
- $\overline{A \cap B} = \overline{A} \cup \overline{B}$

2. **Distributive Laws**

- $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
- $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$

3. **Complementation Law**

- $\overline{\overline{A}} = A$

4. **Commutative Law**

- $A \cup B = B \cup A$
- $A \cap B = B \cap A$

2.3 Section 2.3 - Functions

Definition 19

Let A and B be sets

A **relation** from A to B is a subset of $A \times B$

Example 23

Let $A = \{1, 4, 9\}$ and $B = \{\pm 1, \pm 2, \pm 3\}$

1. $|A \times B| = 3 \cdot 6 = 18$
2. Let R be the relation from A to B defined by

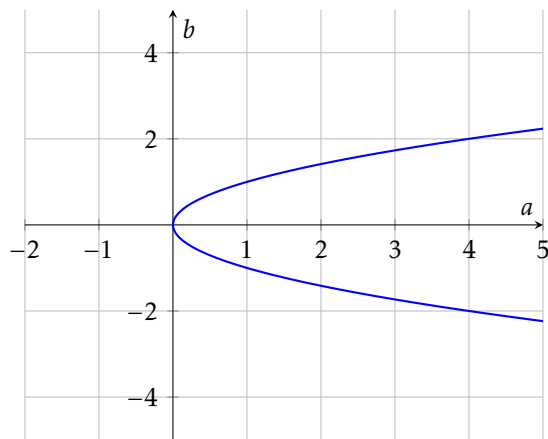
$$R = \{(a, b) \mid a \in A, b \in B, a = b^2\}$$

$$R = \{(1, 1), (1, -1), (4, 2), (4, -2), (9, 3), (9, -3)\}$$

Note

R is a relation but not a function

Why?



This graph of the relation fails the vertical line test, hence it is not a function

There exists an element of A that corresponds to more than one element of B

a	b
1	-1
1	1
4	-2
4	2
9	-3
9	3

Every function is a relation but not all relations are functions

Definition 20 (Function)

Let A and B be sets

A **function** f from A to B , denoted as

$$f : A \rightarrow B$$

is a relation from A to B such that

$$\forall a \in A \exists! b \in B, f(a) = b$$

$\exists!$ means "there exists a **unique**..."

Definition 21 (Domain and Codomain)

We call the set A the **domain** of f and the set B the **codomain**.

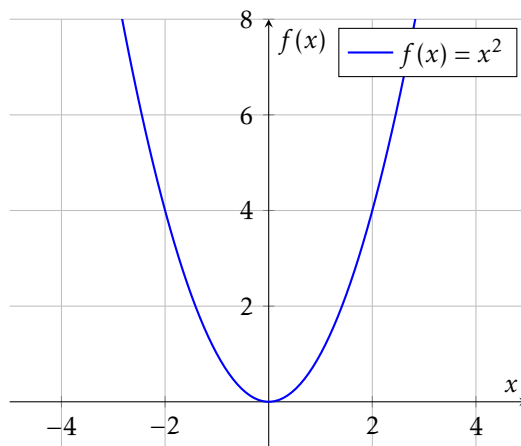
The **range** or **image** of f is the set

$$\text{Im}(f) = \{b \in B \mid \exists a \in A, f(a) = b\}$$

Example 24

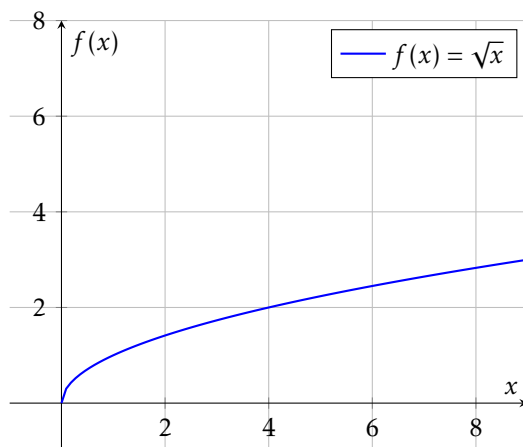
Determine if the following are functions

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = x^2$



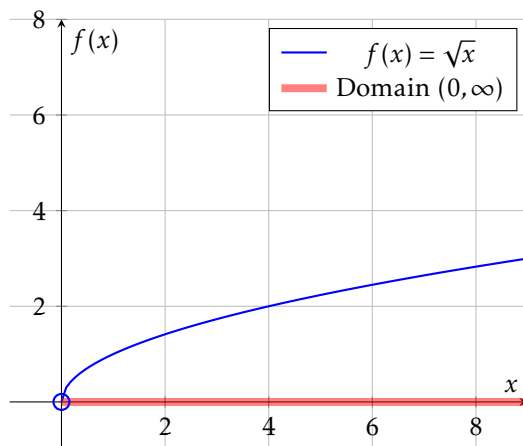
Domain: x axis
 Codomain: y axis
 $\text{Im}(f) = [0, \infty]$
 $\therefore$ Yes

2. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



No since not all inputs are valid

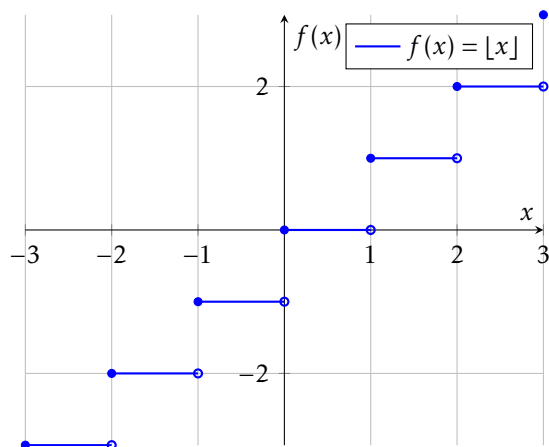
3. $f : \mathbb{R}^+ \rightarrow \mathbb{R}, f(x) = \sqrt{x}$



$$\text{Im}(f) = (0, \infty)$$

Yes since all the inputs have a unique output (the origin is excluded since the domain is $\mathbb{R}^+$)

4. $f : \mathbb{R} \rightarrow \mathbb{Z}, f(x) = \lfloor x \rfloor$ (floor function, aka greatest integer function)

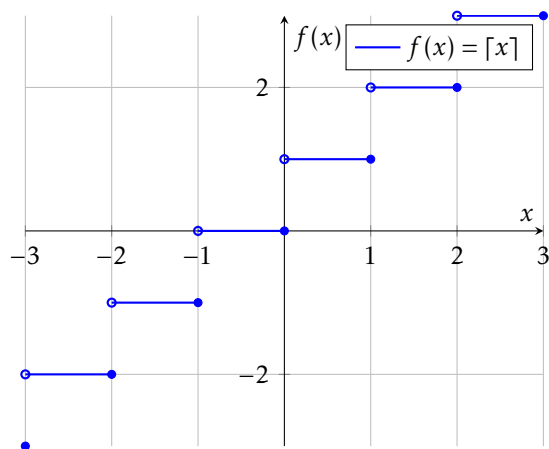


x	$\lfloor x \rfloor$
-1	-1
0	0
0.5	0
0.99	0
1.01	1

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

5. $f : \mathbb{R} \rightarrow \mathbb{Z}$, $f(x) = \lceil x \rceil$ (ceiling function, aka least integer function)



x	$\lfloor x \rfloor$
-1	-1
0	0
0.5	1
0.99	1
1.01	2

$$\text{Im}(f) = \mathbb{Z}$$

All of x axis is domain, but only **integer** parts of y axis is codomain

Definition 22 (Surjective)

Let $f : A \rightarrow B$ be a function in the set A and B

We say f is **onto** or **surjective** if

$$\text{Im}(f) = B$$

or

$$\forall b \in B \exists a \in A, f(a) = b$$

or

All of codomain is covered by the function

Definition 23 (Injective)

We say f is **one to one** or **injective** if

$$\forall b \in B, \text{ the equation } f(a) = b \text{ has at most one solution}$$

Another way of thinking of this is:

Injective if it passes the horizontal line test

or

there is a unique b for all a

or

no two inputs have the same output

or

$$f(a) = f(b) \implies a = b$$

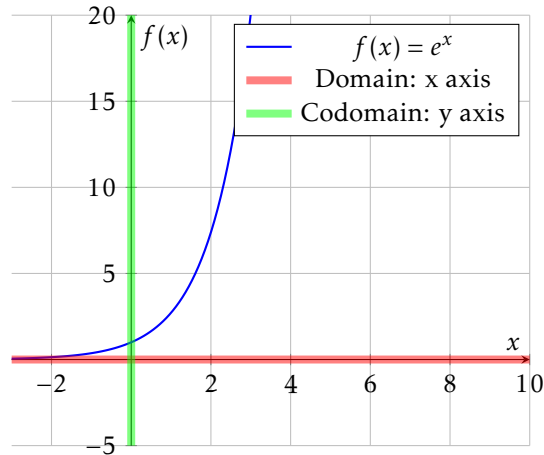
Definition 24 (Bijective)

If f is both **surjective** and **injective**, then we say f is **bijective**

Example 25

Determine if the following functions are injective, surjective, and / or bijective

1. $f : \mathbb{R} \rightarrow \mathbb{R}, f(x) = e^x$

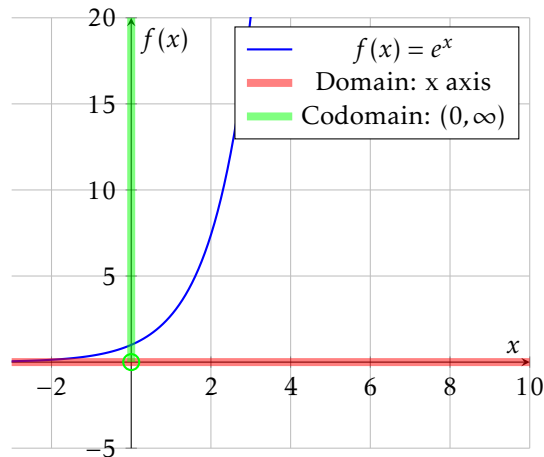


Injective since all x values have a valid output

Not surjective since not all values on the y axis have a corresponding input value

Not bijective since it's not surjective

2. $f : \mathbb{R} \rightarrow \mathbb{R}^+, f(x) = e^x$



Injective since all x values have a valid output

Surjective since all values on the y axis have a corresponding input value

Bijective since it's both surjective and injective

There's more examples, write them down when you get the chance

Definition 25 (Composition)

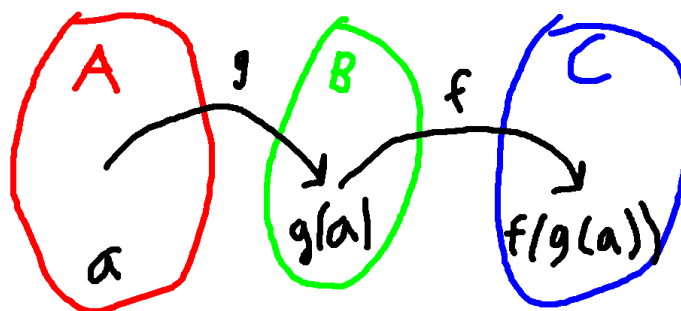
Given the functions $f : B \rightarrow C$ and $g : A \rightarrow B$

Where A, B, C are sets, then we define

$$f \circ g : A \rightarrow C \text{ by}$$

$$(f \circ g)(a) = f(g(a))$$

called the **composition** of f and g



Example 26 (Examples of Compositions)

Let $f(x) = x - 3$, $g(x) = x^3 + 9x^2 + 27x + 27 = (x + 3)^3$

Compute the following

1. $(f \circ g)(x)$

$$\begin{aligned} (f \circ g)(x) &= f(g(x)) \\ &= (x^3 + 9x^2 + 27x + 27) - 3 \\ &= x^3 + 9x^2 + 27x + 24 \end{aligned}$$

$$2. (g \circ f)(x)$$

$$\begin{aligned}(g \circ f)(x) &= g(f(x)) \\ &= (x-3)^3 + 9(x-3)^2 + 27(x-3) + 27 \\ &= [(x-3) + 3]^3 \\ &= x^3\end{aligned}$$

2.4 Section 2.4 - Sequences and Series

Definition 26 (Sequence, Explicit Formula / nth term)

A **sequence** is an ordered list of numbers

$$a_1, a_2, a_3, \dots$$

Equivalently, a sequence is a function $f: \mathbb{Z}^+ \rightarrow \mathbb{R}$

We call f the **explicit formula** or **nth term** of a sequence

Example 27

Determine the next 3 terms of the following sequences and the explicit formula

$$1. \quad 5 \xrightarrow{+7}, 12 \xrightarrow{+7}, 19 \xrightarrow{+7}, 26 \xrightarrow{+7}, \dots, 33 \xrightarrow{+7}, 44 \xrightarrow{+7}$$

Solution:

$$\begin{aligned}a_n &= 7n + b \\ 5 &= a_1 = 7(1) + b \\ b &= -2 \\ \boxed{a_n &= 7n - 2}\end{aligned}$$

$$2. \quad 7 \xrightarrow{\cdot 3}, 21 \xrightarrow{\cdot 3}, 63 \xrightarrow{\cdot 3}, 189 \xrightarrow{\cdot 3}, \dots, 567 \xrightarrow{\cdot 3}, 1701 \xrightarrow{\cdot 3}, 5103 \xrightarrow{\cdot 3}$$

Solution:

$$a_n = c3^n$$

$$a_1 = 7 = c3^1$$

$$c = \frac{7}{3}$$

$$a_n = \frac{7}{3}3^n$$

$$\boxed{a_n = 7 \cdot 3^{n-1}}$$

3. 1, 2, 6, 24, 120, 720, 5040

$\boxed{a_n = n!}$, read as "n factorial"

$$0! = 1$$

4. 1, 1, $\underbrace{2}_{1+1}$, $\underbrace{3}_{2+1}$, $\underbrace{5}_{3+2}$, $\underbrace{8}_{5+3}$, $\underbrace{13}_{8+5}$, $\underbrace{21}_{13+8}$,

Solution:

$$f_n = \frac{1}{\sqrt{5}} \left[\left(\frac{1+\sqrt{5}}{2} \right)^n - \left(\frac{1-\sqrt{5}}{2} \right)^n \right] \quad (\text{We'll derive this later})$$

Another solution is

$$a_1 = 1$$

$$a_2 = 2$$

$$a_n = a_{n-1} + a_{n-2}, n \geq 3$$

Definition 27

A **recursive** sequence is defined in terms of the previous terms of the sequence

Example 28

Here are some examples of recursive sequences

1. $a_1 = 2, a_2 = 3, a_n + 1 = 5a_n - 6a_{n-1}, n \geq 2$

$$\begin{aligned}
 n = 2, a_3 &= 5a_2 - 6a_1 = 5 \cdot 3 - 6 \cdot 2 = 15 - 12 = 3 \\
 n = 3, a_4 &= 5a_3 - 6a_2 = 5 \cdot 3 - 6 \cdot 3 = 15 - 18 = -3 \\
 n = 4, a_5 &= 5a_4 - 6a_3 = 5 \cdot (-3) - 6 \cdot 3 = -15 - 18 = -33
 \end{aligned}$$

$$2. \ a_1 = 1, a_2 = 2, a_3 = 4, a_n = a_{n-1} + a_{n-2} - n - 3, n \geq 4$$

$$a_4 = 4 + 2 - 1 = 5$$

$$a_5 = 5 + 4 - 2 = 7$$

$$a_6 = 7 + 5 - 4 = 8$$

Definition 28 (Series)

A **series** is the sum of an ordered list of numbers

$$a_1 + a_2 + a_3 + \dots$$

$$\sum_{k=1}^{\infty} a_k$$

Where:

k = Index

1 = Starting value

∞ = Ending value

$a_k = k^{\text{th}}$ term

Example 29

Find the sum of the series

$$1 + 2 + 3 + \dots + 100$$

Solution:

$$\begin{aligned} &1 + 2 + 3 + \dots + 98 + 99 + 100 \\ &(1 + 100) + (2 + 99) + (3 + 98) + \dots \\ &(101) + (101) + (101) + \dots \end{aligned}$$

Since there's 100 things we need to add, there's 50 pairs

So

$$\begin{aligned} \text{sum} &= 101(50) \\ &= 5050 \end{aligned}$$

Example 30

But what about in general?

$$1 + 2 + 3 + \dots + n = ?$$

We have two cases:

1. When n is even
2. When n is odd

If we assume n is even, then

$$\boxed{1 + 2 + 3 + \dots + n = (n + 1) \cdot \frac{n}{2}} \quad (2.1)$$

Since $(n + 1)$ gives the value of each pair and $\frac{n}{2}$ gives the number of pairs

If we assume n is odd, then

$$\begin{aligned} [1 + 2 + 3 + \dots + (n - 1)] + n &= n \cdot \frac{n - 1}{2} \\ &= \frac{n^2 + n}{2} \\ &= (n + 1) \cdot \frac{n}{2} \end{aligned}$$

since n is the value of each pair $(n - 1 + 1)$ and $\frac{n-1}{2}$ is the number of pairs

Another way to think about it is to let S represent the sum

$$\begin{array}{r}
 S = 1 + 2 + 3 + \dots + (n-2) + (n-1) + n \\
 S = n + (n-1) + (n-2) + \dots + 3 + 2 + 1 \quad \text{(Switch Order)} \\
 \hline
 2S = \underbrace{(n+1) + (n+1) + (n+1) + \dots + (n+1) + (n+1) + (n+1)}_n \quad \text{(Add)} \\
 \boxed{S = (n+1) \cdot \frac{n}{2}}
 \end{array}$$

Writing this is summation notation

$$\sum_{k=1}^n \frac{n(n+1)}{2}$$

To Review...

1.

$$\sum_{k=1}^n c = \underbrace{c + c + \dots + c}_n = \boxed{nc} \quad \text{(North Carolina Rule)}$$

2.

$$\sum_{k=1}^n k = 1 + 2 + 3 + \dots + n = \boxed{(n+1)\frac{n}{2}} \quad \text{(Pairing Rule)}$$

3.

$$\sum_{k=1}^n k^2 = 1^2 + 2^2 + 3^2 + \dots + n^2 = \boxed{\frac{n(n+1)(2n+1)}{6}} \quad \text{(Prove Later)}$$

4.

$$\sum_{k=1}^n k^3 = 1^3 + 2^3 + 3^3 + \dots + n^3 = \boxed{\left[\frac{n(n+1)}{2}\right]^2} \quad \text{(Prove Later)}$$

You **MUST** know these 4! They will be on exams!

Example 31

Compute the Following

1.

$$\sum_{k=1}^6 k = \frac{6}{2}(6+1) = \boxed{21} \quad (\text{Use formula})$$

2.

$$\begin{aligned} \sum_{k=1}^6 2k &= (2 \cdot 1) + (2 \cdot 2) + (2 \cdot 3) + (2 \cdot 4) + (2 \cdot 5) + (2 \cdot 6) \\ &= 2(1 + 2 + 3 + 4 + 5 + 6) \\ &= 2 \sum_{k=1}^6 k \\ &= 2 \cdot 21 \\ &= \boxed{42} \end{aligned}$$

NoteThis means that we can **factor** in sums!

3.

$$\sum_{k=1}^6 2 = 6 \cdot 2 = \boxed{12} \quad (\text{Use formula})$$

4.

$$\begin{aligned} \sum_{k=1}^6 (k+2) &= (1+2) + (2+2) + (3+2) + (4+2) + (5+2) + (6+2) \\ &= (1+2+3+4+5+6) + (2+2+2+2+2+2) \\ &= \sum_{k=1}^6 k + \sum_{k=1}^6 2 \\ &= 21 + 12 \\ &= \boxed{33} \end{aligned}$$

Note

This means that in general,

$$\sum_{k=1}^n c a_k = c \sum_{k=1}^n a_k \quad (\text{Factor out constants})$$

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k \quad (\text{Split})$$

5.

$$\sum_{k=1}^{200} (k+1)^2 = \sum_{k=1}^{200} (k^2 + 2k + 1) \quad (\text{Expand})$$

$$= \sum_{k=1}^{200} k^2 + 2 \sum_{k=1}^{200} k + \sum_{k=1}^{200} 1 \quad (\text{Split up})$$

$$= \frac{200(201)(401)}{6} + 2 \cdot \frac{200(201)}{2} + 200 \cdot 1 \quad (\text{Use formulas})$$

$$\boxed{= 2727100} \quad (\text{Use Calculator})$$

We could also do

$$\sum_{k=1}^{200} (k+1)^2 = 2^2 + 3^2 + 4^2 + \dots + 201^2$$

$$= (1^2 + 2^2 + 3^2 + 4^2 + \dots + 201^2) - 1^2$$

$$= \frac{201(202)(403)}{2} - 1$$

$$\boxed{= 2727100} \quad (\text{Use Calculator})$$

Theorem 1 (Constant Shift)

This means that when you have a **constant shift** like $(k+1)$ is a shift from k , then

$$\sum_{k=1}^n f(k+a) = \sum_{k=1}^{n+a} f(k) - \sum_{k=1}^a f(k)$$

or

$$\sum_{k=m}^n a_k = \sum_{k=1}^n a_k - \sum_{k=1}^{m-1} a_k$$

Where $f(k)$ is a function like k^2 that has it's own formula for the sum like

$$\sum_{k=1}^n f(k) = \sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$$

Using this on the other example,

$$\begin{aligned} \sum_{k=1}^{200} (k+1)^2 &= \sum_{k=1}^{201} k^2 - \sum_{k=1}^1 k^2 \\ &= \frac{201(202)(403)}{2} - \frac{1(2)(1)}{2} \\ &= 2727101 - 1 \\ &= 2727100 \end{aligned} \quad \text{(Use Calculator)}$$

6.

$$\sum_{k=1}^{10} 2^k = S$$

$$S = 2^1 + 2^2 + 2^3 + \dots + 2^9 + 2^{10}$$

$$2S = 2^2 + 2^3 + 2^4 + \dots + 2^{10} + 2^{11}$$

(Subtract)

$$S - 2S = 2^1 - 2^{11}$$

$$S(1 - 2) = 2^1 - 2^{11}$$

$$S = \frac{2 - 2^{11}}{1 - 2} = \frac{2 - 2048}{-1}$$

$$\boxed{= 2046}$$

Note

This means that

$$S = \sum_{k=0}^n r^k$$

(When k starts at 0)

$$S = 1 + r^1 + r^2 + \dots + r^{n-1} + r^n$$

$$rS = 0 + r^1 + r^2 + r^3 + \dots + r^{n-1} + r^n + r^{n+1}$$

(Subtract)

$$S - rS = 1 - r^{n+1}$$

$$S(1 - r) = 1 - r^{n+1}$$

$$\boxed{S = \frac{1 - r^{n+1}}{1 - r}}$$

$$S = \sum_{k=1}^n r^k \quad (\text{When } k \text{ starts at } 1)$$

$$\begin{aligned} S &= r^1 + r^2 + \dots + r^{n-1} + r^n \\ &= r(1 + r^1 + r^2 + \dots + r^{n-2} + r^{n-1}) \\ &= r \sum_{k=1}^{n-1} r^k \\ &= r \cdot \frac{1 - r^{n-1+1}}{1 - r} \\ &= r \cdot \frac{1 - r^n}{1 - r} \end{aligned}$$

In general,

$$\sum_{k=\text{start}}^n r^k = \frac{r^{\text{start}} - r^{n+1}}{1 - r}$$

Note

This works for $r \in \mathbb{R}, (-\infty, \infty)$, including **negative** and **fractions**!

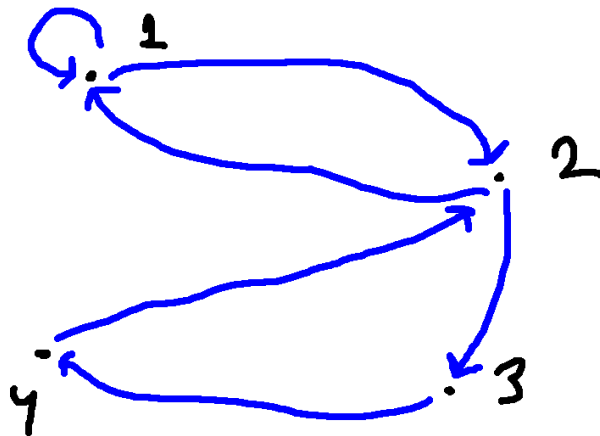
This can also work for sums going to ∞ , remember that a fraction to ∞ is just 0

There's more examples, but I'm too lazy to write them, so I'll just write the equations

Note

$$\sum_{k=0}^{\infty} k r^k = \sum_{k=1}^{\infty} k r^k = \boxed{\frac{r}{(1-r)^2}}$$

2.6 Section 2.6 - Matrices



Definition 29 (Directed Graph)

This is a **directed graph** and could be used to model:

- How webpages are linked
- Flights between cities
- Who knows who

How can a computer store this directed graph?

Definition 30 (Matrix)

We will use a **matrix**, which is a rectangular array of numbers

For this network, we define a matrix

$$A = [a_{ij}]$$

Where

$$a_{ij} = \begin{cases} 0 & \text{if web page } i \text{ doesn't link to } j \\ 1 & \text{if web page } i \text{ does link to } j \end{cases}$$

i = row (starts at 1)
 j = column (starts at 1)

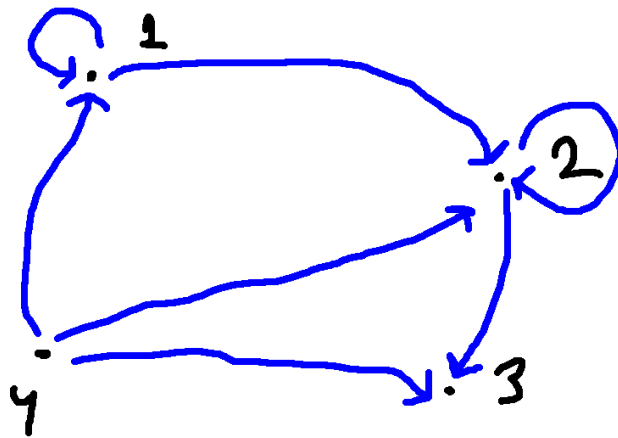
The resulting matrix would look like

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \\ a_{41} & a_{42} & a_{43} & a_{44} \end{bmatrix}$$

For this network, the matrix is

$$A = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix}$$

Let's modify the graph now



This matrix is

$$B = \begin{bmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

Matrix Addition and Subtraction

We define the addition of matrices by adding corresponding components

$$A + B = \begin{bmatrix} 2 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

The same can be said for Subtraction

$$A - B = \begin{bmatrix} 0 & 0 & 1 & 1 \\ 1 & -1 & 1 & -1 \\ 0 & 0 & 0 & 1 \\ -1 & 1 & -1 & 1 \end{bmatrix}$$

Note

Two matrices can only be added/subtracted if they're the same size

Question

Going back to the first network, how many ways can we start at 1 and return to 1 after passing through exactly 1 webpage?

$$1 \rightarrow 1 \rightarrow 1$$

$$1 \rightarrow 2 \rightarrow 1$$

$$a_{12} \rightarrow a_{21}$$

$$\begin{bmatrix} a_{11} & a_{12} & 0 & 0 \\ 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \begin{bmatrix} a_{11} & 1 & 0 & 0 \\ 1 & 0 & 1 & 0 \\ a_{21} & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 1 & 1 & 0 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Example 32

Let

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}, B = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix}, C = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

1. $A + B$

Solution:

Undefined, due to dimensions

2. $6A$

Solution:

$$6A = \begin{bmatrix} 6 & 12 & 18 \\ 24 & 30 & 36 \end{bmatrix}$$

3. AB

Solution:

$$AB = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} -1 & 5 \\ -1 & 11 \end{bmatrix}$$

4. BA

Solution:

$$BA = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 3 & 3 & 3 \\ 4 & 5 & 6 \end{bmatrix}$$

5. BC

Solution:

$$BC = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 2 & 4 \end{bmatrix}$$

6. CB

Solution:

Undefined, due to dimensions

7. $C^2 = CC$

$$C^2 = CC = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix} = \begin{bmatrix} 5 & 10 \\ 10 & 20 \end{bmatrix} \\ = 5C$$

8. C^{11}

$$C^2 = 5C$$

$$C^3 = C^2 \cdot C$$

$$= 5C \cdot C$$

$$= 5C^2$$

$$= 5^2 C$$

$$C^{11} = 5^{10} C = 5^{10} \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

Okay let's now define some properties

2.6.1 Properties of Matrices

Definition 31 (Additive Identity)

We define the **Additive Identity** for matrices as

$$A_{n \times m} + \vec{0} = A_{n \times m}, \vec{0} = \begin{bmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}$$

Definition 32 (Multiplicative Identity)

We define the **Multiplicative Identity** for matrices as

$${}_{n \times m} A \cdot I_n = A, I_n = \begin{bmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{bmatrix}$$

But A must be **square**!

Some matrices have a multiplicative inverse such that

Definition 33 (Multiplicative Inverse)

We define the **Multiplicative Inverse** of matrices as

$$A \cdot B = I$$

But only **some** matrices have them, not all do!

Also, A must be square to even qualify for possibly having one

For this class, we define

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Example 33

We have

$$\begin{bmatrix} 4 & 3 \\ 7 & 5 \end{bmatrix}^{-1} = \frac{1}{20 - 21} \begin{bmatrix} 5 & -3 \\ -7 & 4 \end{bmatrix}$$

Chapter 4

Section 4 - Number Theory

4.1 Section 4.1 - Divisibility

Question

When is an integer divisible by 2?

Solution:

When an integer ends in 0, 2, 4, 6, 8

Let's prove this after we develop some notation

Definition 34 (Divides)

For integers $a, b, \in \mathbb{Z}$, we say "a divides b" if b is divisibly by a and we write

$$a \mid b \quad (\text{TRUE value})$$

$$a \nmid b \quad (\text{FALSE value})$$

Example 34

$2 \mid 48$ since 48 is divisible by 2

but $7 \nmid 48$ since 48 is not divisible by 7

Let's prove

for $n \in \mathbb{Z}, 2 \mid n$ if $\exists k \in \mathbb{Z}$ such that $n = 10k$ or $10k + 2$ or $10k + 4$ or $10k + 6$ or $10k + 8$

We could also say

$$\exists k \in \mathbb{Z} \text{ such that } n = 10k + r \text{ where} \\ r = \{0, 2, 4, 6, 8\}$$

Proof

(Proof of the converse)

Let $n \in \text{set } \mathbb{Z}$

Then base 10 representation of n is of the form

$$n = d_k 10^k + d_{k-1} 10^{k-1} + \dots + d_1 10 + d_0$$

Since $2 \mid n$, then $n = 2m$ for some $m \in \mathbb{Z}$

We have

$$n = 10(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) + d_0 \\ n = 2 \left[5(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) \right] + d_0$$

Solving for d_0 we get

$$d_0 = n - 2 \left[5(d_k 10^{k-1} + d_{k-1} 10^{k-2} + \dots + d_1) \right]$$

Since d_0 is the sum of two even numbers, then d_0 is even and $d_0 \in \{0, 2, 4, 6, 8\}$

Let's prove some of the properties of the "divides" symbol

In particular, let's rewrite what $a \mid b$ means for $a, b \in \mathbb{Z}$

So, $a \mid b$ means

$$\exists k \in \mathbb{Z} \text{ such that } b = ak$$

We believe if two integers are divisible by an integer, then their sum is divisible by that integer

We prove $a, b, c \in \mathbb{Z}, a \mid b, a \mid c, \implies a \mid (b + c)$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$a \mid c \implies \exists n \in \mathbb{Z} \text{ such that } c = an$$

Then

$$b + c = ak + an = a(k + n)$$

$$\therefore a \mid (b + c) \text{ as desired}$$

Note

This also works for **subtraction**

Proposition 2 (Modulus Multiplication)

$$a, b, c \in \mathbb{Z}, a \mid b, a \mid c \implies a^2 \mid bc$$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$a \mid c \implies \exists n \in \mathbb{Z} \text{ such that } c = an$$

Then

$$bc = (ak)(an)$$

$$= a^2(kn), \text{ proving } a^2 \mid bc \text{ as desired}$$

Proposition 3 (Constant Factor)

$$a, b, c \in \mathbb{Z}, a \mid b \implies a \mid bc$$

Proof

$$a \mid b \implies \exists k \in \mathbb{Z} \text{ such that } b = ak$$

$$bc = akc = a(kc), \text{ implying } a \mid bc, \text{ as desired}$$

Let's rewrite the old proof from earlier, this time proving the actual proposition rather than contrapositive

Proposition 4

For $n \in \mathbb{Z}$, where $n = 10k + r, r \in \{0, 2, 4, 6, 8\}$, then $2 \mid n$

Proof

$$\forall r \in \{0, 2, 4, 6, 8\}, \exists m \in \mathbb{Z} \text{ such that } r = 2m$$

Subbing this in

$$\begin{aligned} n &= 10k + r = 10k + 2m \\ &= 2(5k + m), \text{ implying } 2 \mid n, \text{ as desired} \end{aligned}$$

Proposition 5

For $n \in \mathbb{Z}$, if the sum of the digits is divisible by 3, then the integer n is divisible by 3

Proof.

$$\begin{aligned} n &= d_k 10^k + d_{k-1} 10^{k-1} + \dots + d_1 10 + d_0 \\ n &= (10^k - 1)d_k + (10^{k-1} - 1)d_{k-1} + \dots + 99d_2 + 9d_1 + \underbrace{(d_k + d_{k-1} + \dots + d_2 + d_1 + d_0)}_{\text{sum of digits}} \end{aligned}$$

Because 10^k and so on will always be a bunch of 9's, we see that

$$3 \mid (10^k - 1)d_k + (10^{k-1} - 1)d_{k-1} + \dots + 99d_2 + 9d_1$$

And

$$3 \mid (d_k + d_{k-1} + \dots + d_2 + d_1 + d_0) \text{ by hypothesis}$$

$\therefore n$ is the sum of 2 multiples of 3, proving that $3 \mid n$ as desired

□

Here's just a table for divisibility rules

n is divisible by	if
2	last digit is even
3	sum of digits is divisible by 3
4	last 2 digits is divisible by 4
5	last digit is 0 or 5
6	divisible by both 2 and 3
7	He showed us but I forgot
8	last 3 digits divisible by 8
9	sum of digits is divisible by 9
10	last digit is 0

Definition 35 (Division Algorithm)

The **division algorithm** is a way of expressing division with integers

$$\forall a \in \mathbb{Z} \forall d \in \mathbb{Z}^+ \exists q \in \mathbb{Z} \exists r \in \{0, 1, 2, 3, \dots, d-1\}, a = dq + r$$

Where

d : divisor
 q : quotient
 r : remainder

Example 35

Let $a = 341$ and $d = 7$

$$341 = \underbrace{7}_{\text{divisor}} \cdot \underbrace{48}_{\text{quotient}} + \underbrace{5}_{\text{remainder}}$$

However we can write this in a more compact way

$$341 \pmod{\underbrace{7}_{\text{divisor}}} = \underbrace{5}_{\text{remainder}}$$

Example 36

Let's give examples of division, we're practically finding the remainder

1.

$$\frac{6568}{9} = 729.777$$
$$6568 - 9 \cdot 729 = 7$$

2.

$$\frac{343348}{7} = 49049.714$$
$$343348 - 7 \cdot 49049 = 5$$

3.

$$\frac{-100}{13} = -7.6923$$
$$-100 - 13(-7) = -9$$
$$= -100 - 13(-8) = 4$$
$$-100 = (13)(-8) + 4$$

Note

For negatives, round down to the next negative integer in order to get a positive remainder

Example 37

Let's do more examples but with mod

1.

$$1023 \pmod{11} = 0$$

2.

$$6568 \pmod{9} = 7$$

3.

$$343348 \pmod{7} = 5$$

4.

$$-100 \pmod{13} = 4$$

Definition 36 (Congruence)

Let $a, b \in \mathbb{Z}$ and $m \in \mathbb{Z}^+$

We say a is **congruent** to b modulo m if

$$a \equiv b \pmod{m}$$

This is basically saying are the remainders the same

Written as

$$a \equiv b \pmod{m}$$

And if they're not then

$$a \not\equiv b \pmod{m}$$

Note

If $a \equiv b \pmod{m}$ then by **Division Algorithm**

$$a = mq_a + r \text{ and } b = mq_b + r$$

Subtrating the equations we get

$$\begin{aligned} a - b &= (mq_a + r) - (mq_b + r) \\ &= m(q_a - q_b) \\ \boxed{\therefore m \mid (a - b)} \end{aligned}$$

Proposition 6

Let $a, b, c, d \in \mathbb{Z}, m \in \mathbb{Z}^+$

Suppose $a \equiv b \pmod{m}$ and $c \equiv d \pmod{m}$

1. $a + c \equiv b + d \pmod{m}$

Proof

$$\begin{aligned} a &\equiv b \pmod{m} \implies m \mid (a - b) \\ c &\equiv d \pmod{m} \implies m \mid (c - d) \end{aligned}$$

Then

$$\begin{aligned}m &\mid [(a-b) + (c-d)] \\m &\mid [(a+c) - (b+d)] \\ \boxed{a+c \equiv b+d \pmod{m}} &\text{ as desired}\end{aligned}$$

2. $ac \equiv bd \pmod{m}$

Proof

$$\begin{aligned}a \equiv b \pmod{m} &\implies m \mid (a-b) \\c \equiv d \pmod{m} &\implies m \mid (c-d)\end{aligned}$$

$$\begin{aligned}m \mid (a-b) &\implies \exists k_1 \in \mathbb{Z} \text{ such that } a-b = mk_1 \\m \mid (c-d) &\implies \exists k_2 \in \mathbb{Z} \text{ such that } c-d = mk_2\end{aligned}$$

Then $a = b + mk_1$ and $c = d + mk_2$

$$\begin{aligned}ac &= (b + mk_1)(d + mk_2) && \text{(FOIL)} \\&= bd + bmk_2 + dm k_1 + m^2 k_1 k_2\end{aligned}$$

Moving bd

$$\begin{aligned}ac - bd &= bmk_2 + dm k_1 + m^2 k_1 k_2 \\&= m(bk_2 + dk_1 + mk_1 k_2) \\ \boxed{\therefore m \mid (ac - bd)}\end{aligned}$$

3. if $h \in \mathbb{Z}$ then $ah \equiv bh \pmod{m}$

Proof

$$a \equiv b \pmod{m} \implies m \mid (a - b)$$

Then since $h \in \mathbb{Z}$, we have

$$m \mid h(a - b)$$

$$m \mid (ah - bh)$$

$$\boxed{ah \equiv bh \pmod{m}}$$

4.2 Section 4.2 - Binary

Let's compute

$$67^2 025 \pmod{77}$$

Work

Let's see if FLT works

$77 = 7 \cdot 11$ is **NOT** prime, so FLT does not apply

Let's see if EFLT works

1. $77 = 7 \cdot 11$ is the product of 2 distinct primes
2. $\gcd(67, 77) = 1$

So EFLT applies

$$6^{(7-1)(11-1)} \equiv 1 \pmod{77}$$

$$67^{60} \equiv 1 \pmod{77}$$

Breaking up 2025...

$$2025 = 33 \cdot 60 + 45$$

$$67^{2025} \equiv 67^{45} \pmod{77}$$

Break up 45 into powers of 2

$$\begin{aligned}
 67 & \pmod{77} = 67 \\
 67^2 & \pmod{77} = 23 \\
 67^4 & \pmod{77} = 23^2 \pmod{77} \\
 & = 67 \\
 67^8 & \pmod{77} = 23 \\
 67^{16} & \pmod{77} = 67 \\
 67^{32} & \pmod{77} = 23
 \end{aligned}$$

$$\begin{aligned}
 67^{2025} & \equiv 67^{45} \pmod{77} \\
 & \equiv 67^{32+8+4+1} \pmod{77} \\
 & \equiv 67^{32} \cdot 67^8 \cdot 67^4 \cdot 67^1 \pmod{77} \\
 & \equiv 23 \cdot 23 \cdot 67 \cdot 67 \cdot 67 \\
 & \equiv 23^2 \cdot 67^2 \pmod{77} \\
 & \boxed{\equiv 1} \pmod{77}
 \end{aligned}$$

How do we express 2025 as powers of 2?

Let's list the powers of 2

$$\begin{aligned}
 2^0 & = 1 \\
 2^1 & = 2 \\
 2^2 & = 4 \\
 2^3 & = 8 \\
 2^4 & = 16 \\
 2^5 & = 32 \\
 & \vdots \\
 2^{10} & = 1024 \\
 2^{11} & = 2048
 \end{aligned}$$

Now let's break up 2025 using a method similar to the Euclidean Algorithm

$$2025 = 2^{10} + 1001$$

$$1001 = 2^9 + 489$$

$$489 = 2^8 + 233$$

$$233 = 2^7 + 105$$

$$105 = 2^6 + 41$$

$$41 = 2^5 + 9$$

$$9 = 2^3 + 2^0$$

$$2025 = 2^{10} + 2^9 + 2^8 + 2^7 + 2^6 + 2^5 + 2^3 + 2^0$$

We can use this to write it in **binary representation**.

$$\underbrace{1}_{2^{10}} \underbrace{1}_{2^9} \underbrace{1}_{2^8} \underbrace{1}_{2^7} \underbrace{1}_{2^6} \underbrace{1}_{2^5} \underbrace{0}_{2^4} \underbrace{1}_{2^3} \underbrace{0}_{2^2} \underbrace{0}_{2^1} \underbrace{1}_{{2^0}_2}$$

Note

The 2 subscript denotes that this is in binary / base 2

Where 1 and 0 represent the coefficient attached to each power. Since it's 1 and 0, it simply represents the power being present and 0 being absent

But notice that if the number n is odd, then the last number of the binary representation is 1

Else if n is even then the last digit will be 0

In other words

$$n \bmod 2 = 1 \text{ if odd}$$

$$n \bmod 2 = 0 \text{ if even}$$

$$\begin{aligned}
2025 &= 2 \cdot 1012 + 1 \\
1012 &= 2 \cdot 506 + 0 \\
506 &= 2 \cdot 253 + 0 \\
253 &= 2 \cdot 126 + 1 \\
126 &= 2 \cdot 63 + 0 \\
63 &= 2 \cdot 31 + 1 \\
31 &= 2 \cdot 15 + 1 \\
15 &= 2 \cdot 7 + 1 \\
7 &= 2 \cdot 3 + 1 \\
3 &= 2 \cdot 1 + 1 \\
1 &= 2 \cdot 0 + 1
\end{aligned}$$

Writing the sequence of remainders backwards we see

$$\underbrace{1}_{\text{last}} \ 1 \ 1 \ 1 \ 1 \ 1 \ 0 \ 1 \ 0 \ 0 \ \underbrace{1_2}_{\text{first}} = 2025$$

But why does this work?

Given $n \in \mathbb{Z}$

Want $n = a_k 2^k + a_{k-1} 2^{k-1} + \dots + a_2 2^2 + a_1 2 + a_0$

Using the division algorithm,

$$n = 2q_0 + a_0$$

Then

$$q_0 = \frac{n - a_0}{2}$$

Now repeat this with q_0

$$\begin{aligned}
q_0 &= 2q_1 + a_1 \\
&\vdots
\end{aligned}$$

Let's do an example.

Example 38

Write 1577 in binary

$$1577 = 2 \cdot 788 + 1$$

$$788 = 2 \cdot 394 + 0$$

$$394 = 2 \cdot 197 + 0$$

$$197 = 2 \cdot 98 + 1$$

$$98 = 2 \cdot 49 + 0$$

$$49 = 2 \cdot 24 + 1$$

$$24 = 2 \cdot 12 + 0$$

$$12 = 2 \cdot 6 + 0$$

$$6 = 2 \cdot 3 + 0$$

$$3 = 2 \cdot 1 + 1$$

$$1 = 2 \cdot 0 + 1$$

Writing the bottom to top left to right...

11000101001

We could check this by just doing

$$1577 = 2^{10} + 2^9 + \dots$$

4.2.1 Base n Representation

Y'know, 2 isn't special. We could actually use any number!

Let's try representing 1577 in base 8. Remember that each remainder we get is the coefficient to the corresponding power

$$\begin{aligned}
 1577 &= 8 \cdot 197 + \boxed{1} \\
 197 &= 8 \cdot 24 + \boxed{5} \\
 24 &= 8 \cdot 3 + \boxed{0} \\
 3 &= 8 \cdot 0 + \boxed{3}
 \end{aligned}$$

Rewriting this, we get

$$\underbrace{3}_{8^3} \underbrace{0}_{8^3} \underbrace{5}_{8^2} \underbrace{1}_{8^0}$$

So

$$1577 = 3 \cdot 8^3 + 5 \cdot 8^1 + 1 \cdot 8^0$$

Now let's do it with base 16

$$\begin{aligned}
 1577 &= 16 \cdot 98 + \boxed{9} \\
 98 &= 16 \cdot 6 + \boxed{2} \\
 6 &= 16 \cdot 0 + \boxed{6}
 \end{aligned}$$

$$1577 = 629_{16}$$

$$1577 = 6 \cdot 16^2 + 2 \cdot 16^1 + 9 \cdot 16^0$$

For our last example, write the $AB0CD_{16}$ in base 10 (aka normal numbers)

Recall that

$$\begin{aligned}
 A &= 10 \\
 B &= 11 \\
 &\vdots
 \end{aligned}$$

AB0CD is just 10 11 0 12 13

Since there's 5 "numbers" that means we start with $16^{5-1} = 16^4$

$$\begin{aligned} AB0CD_{16} &= 10 \cdot 16^4 + 11 \cdot 16^3 + 0 \cdot 16^2 + 12 \cdot 16^1 + 13 \cdot 16^0 \\ &= \boxed{700621} \end{aligned} \quad \text{(just plug into calc)}$$

4.3 Section 4.3 - Primes

Definition 37

A **prime number** is a positive integer with exactly 2 **distinct factors**

Example 39

Here are the first few primes

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, ...

is 257 prime?

We use **trial division**

We'll use a loop

```
for k = 2 to 256
  if 257 % k = 0
    then return "composite"
  else
    k += 1
```

For efficiency, we could just go up to $\sqrt{257}$, or $\lfloor \sqrt{256} \rfloor$

Second method: **Sieve of Eratostheres**

1	2	3	4	5	6	7	8	9	10
11	12	13	14	15	16	17	18	19	20
21	22	23	24	25	26	27	28	29	30
31	32	33	34	35	36	37	38	39	40
41	42	43	44	45	46	47	48	49	50

I honestly forgot how this works since I didn't write it down too well

Third method: **Fermat's Method**

if $n = a^2 - b^2$

then $n = (a - b)(a + b)$

Check: $a - b > 1$ and $a + b > 1$

If neither is 1, then n is composite. If both are greater than 1, then they can't be n .

Fourth Method: **Quadratic Sieve**

Solve

$$a^2 - b^2 \equiv 0 \pmod{n}$$

Then

$$n \mid a^2 - b^2 \implies a^2 - b^2 = nk, k \in \mathbb{Z}$$

Proposition 7

There are infinitely primes

Proof

Suppose on the contrary, there are finitely many primes such that all primes can be written like

$$p_1, p_2, \dots, p_k$$

Consider

$$N = p_1 p_2 \cdots p_k + 1$$

Case 1: Suppose N is prime

That means that

N is larger than p_i for $i = 1, 2, \dots, k$

Hence $N \neq p_i$ for $i = 1, 2, \dots, k$. We have a contradiction of our finitely many primes!

Case 2: N is composite

Then N must have a prime factor in the list $p_1, p_2, \dots, p_k$

Suppose p_j is a prime factor of N

Then $p_j \mid N$ and not $p_j \mid p_1 p_2 \cdots p_k$

Then we conclude

$$p_j \mid (N - p_1 p_2 \cdots p_k)$$

Therefore

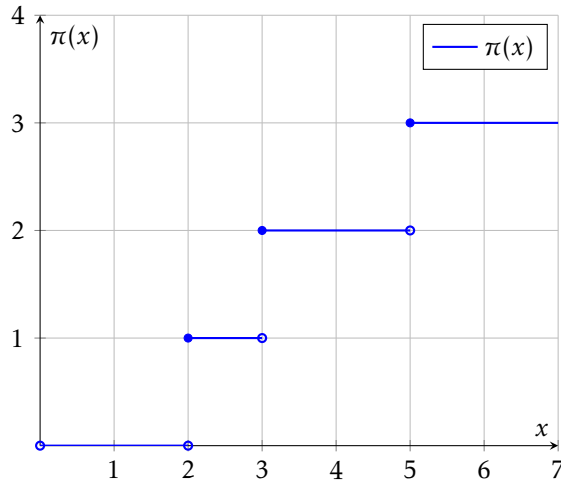
$$p_j \mid 1, \text{ is a contradiction}$$

We conclude there are infinitely primes, as desired (by contradiction)

Definition 38 (Prime Counting Function (Pi))

The function $\pi : \mathbb{R}^+ \rightarrow \mathbb{Z}^+$ is defined by

$\pi(x)$ = number of primes less than or equal to x



Theorem 2 (The Prime Number Theorem)

This theorem is defined that $\pi(x)$ is approximately equal to $\frac{x}{\ln(x)}$ as $x \rightarrow \infty$

$$\lim_{x \rightarrow \infty} \frac{\pi(x)}{\frac{x}{\ln(x)}} = 1$$

In other words,

$$\pi(x) \approx \frac{x}{\ln(x)}$$

Note

This was proven independently by Jacques Hadamard and Charles Jean de la Vallée Poussin in 1896

To recap, we know so far for $a, b, c, d \in \mathbb{Z}, m \in \mathbb{Z}^+$ that

1. $a \equiv b \pmod{m} \iff m \mid (a - b)$
2. $a \equiv b \pmod{m}, c \equiv d \pmod{m}$

- (i) $a + c \equiv b + d \pmod{m}$
- (ii) $ac \equiv bd \pmod{m}$
- (iii) $ac \equiv bc \pmod{m}$

But we see that we can multiply and add with $\equiv$, but how can we divide with modular arithmetic?

Let's go back to algebra

How do we solve this? We need a way to get rid of the 7

$$\begin{aligned}
 7x &= 14 \\
 \frac{7}{7}x &= \frac{14}{7} \\
 1 \cdot x &= 2 \\
 \boxed{x = 2}
 \end{aligned}$$

In this case, we used the **multiplicative inverse** of 7, 7^{-1} , which is $\frac{1}{7}$

$$\begin{aligned}
 7x &= 14 \\
 7^{-1}7x &= 7^{-1} \cdot 14 && \text{(Multiplicative Inverse)} \\
 1x &= 2 && \text{(Multiplicative Identity)} \\
 x &= 2
 \end{aligned}$$

In order to divide, mod m , we need to understand **multiplicative inverse**

1 is still the multiplicative identity, so given $a \in \mathbb{Z}$, the multiplicative inverse of a (which we'll call x) satisfies

$$ax \equiv 1 \pmod{m}$$

We conclude that

$$m \mid (ax - 1)$$

This means that

$$\begin{aligned} \exists k \in \mathbb{Z} \text{ such that} \\ ax - 1 &= mk \\ \text{and} \\ ax - mk &= 1 \end{aligned}$$

Our first task is: when does this equation have a solution?

Claim 1

If $m \mid a$ then $ax - mk = 1$ has no integer solutions

Proof

If $m \mid a$ then $\exists n \in \mathbb{Z}$ such that $a = nm$

The equation $ax - mk = 1$ becomes

$$\begin{aligned} ax - mk &= 1 \\ nm - mk &= 1 && \text{(Substitute)} \\ m(n - k) &= 1 && \text{(Factor m)} \end{aligned}$$

We conclude that $m \mid 1$, which is not possible

So $ax - mk = 1$ has no integer solutions

Example 40

Let's do an example

Let $m = 12, a = 7$

We want to find the multiplicative inverse of 7 modulo 12

We must solve

$$7x \equiv 1 \pmod{12}$$

Method 1: Brute Force

$$\begin{aligned}
7 \cdot 1 &\pmod{12} = 7 \\
7 \cdot 2 &\pmod{12} = 2 \\
7 \cdot 3 &\pmod{12} = 9 \\
7 \cdot 4 &\pmod{12} = 4 \\
7 \cdot 5 &\pmod{12} = 11 \\
7 \cdot 6 &\pmod{12} = 6 \\
7 \cdot 7 &\pmod{12} = 1
\end{aligned}$$

$$\boxed{7^{-1} \equiv 7 \pmod{12}}$$

This wasn't too long, but it can be very tedious and very long as we work with higher numbers. This works, but it's not good.

Method 2: Adding 0

Since $0 \equiv 12 \pmod{12}$, we can keep on adding 12 to the right side until it's divisible by 7

$$\begin{array}{r}
7x \equiv 1 \pmod{12} \\
+0 \equiv 48 \pmod{12} \\
\hline
7x \equiv 49 \pmod{12} \\
7^{-1}7x \equiv 7^{-1}7 \cdot 7 \pmod{12} \\
\boxed{x \equiv 7 \pmod{12}}
\end{array}$$

This was pretty short, but we got lucky. Normally this would take a lot more steps / additions

Method 3: Euclidean Algorithm

This is the one we'll be using most in this class

Let's rewrite

$$7x \equiv 1 \pmod{12}$$

We can rewrite it as divides

$$12 \mid (7x - 1)$$

Since 12 divides $(7x - 1)$, that means $(7x - 1)$ is a multiple of 12, so

$$\begin{aligned} 7x - 1 &= 12k \\ 7x - 12k &= 1 \end{aligned}$$

Now we just repeat the **division algorithm**

We start by breaking up 12

$$\begin{aligned} 12 &= 1 \cdot 7 + 5 \\ 7 &= 1 \cdot 5 + 2 \\ 5 &= 2 \cdot 2 + 1 \end{aligned}$$

Once we have the remainder as 1, we must work backwards

From the equations above, let's solve for each remainder

$$\begin{aligned} 5 &= 2 \cdot 2 + 1 \\ 1 &= 5 - 2 \cdot 2 \end{aligned} \quad \text{(Let's solve for 1)}$$

$$\begin{aligned} 7 &= 1 \cdot 5 + 2 \\ 2 &= 7 - 1 \cdot 5 \end{aligned} \quad \text{(Solve for 2)}$$

$$\begin{aligned} 12 &= 1 \cdot 7 + 5 \\ 5 &= 12 - 1 \cdot 7 \end{aligned} \quad \text{(Solve for 5)}$$

Now, using $1 = 5 - 2 \cdot 2$ as our base, let's repeatedly substitute the remainders

For each line, substitute the divisor then simplify

$$\begin{aligned}
1 &= 5 - 2 \cdot 2 && \text{(now substitute 2)} \\
1 &= 5 - 2(7 - 1 \cdot 5) && \text{(now simplify)} \\
1 &= 5 - 2 \cdot 7 + 2 \cdot 5 \\
1 &= 3 \cdot 5 - 2 \cdot 7 && \text{(now substitute 5)} \\
1 &= 3(12 - 1 \cdot 7) - 2 \cdot 7 && \text{(now simplify)} \\
1 &= 3 \cdot 12 - 3 \cdot 7 - 2 \cdot 7 \\
1 &= 3 \cdot 12 - 5 \cdot 7
\end{aligned}$$

Remember that we were trying to solve for $7x - 12k = 1$, so we conclude that

$$\begin{aligned}
7x - 12k &= 1 \\
7(-5) - 12(-3) &= 1 \\
x = -5, k &= 3
\end{aligned}$$

We conclude that

$$\begin{aligned}
x &\equiv -5 \pmod{12} \\
+0 &\equiv 12 \pmod{12}
\end{aligned}$$

$$x \equiv 7 \pmod{12}$$

4.3.1 Euclidean Algorithm

Definition 39 (Euclidean Algorithm)

Because that last example could be hard to read, I'll rewrite the Euclidean Algorithm but with variables

Let's say we're trying to find the multiplicative inverse of

$$a \pmod{m}$$

We essentially solve

$$a \cdot a^{-1} - mk = 1, \text{ k doesn't matter}$$

Using the Euclidean Algorithm, solve the following until the remainder becomes 1

If the remainder equals 0, that means that the number does not have a multiplicative inverse mod m

$$\begin{array}{ll}
 \boxed{m} = k_1 \cdot \boxed{a} + \boxed{r_1} & \text{(first iteration)} \\
 \boxed{a} = k_2 \cdot \boxed{r_1} + \boxed{r_2} & \text{(second iteration)} \\
 \boxed{r_1} = k_3 \cdot \boxed{r_2} + \boxed{r_3} & \text{(third iteration)} \\
 \boxed{r_2} = k_4 \cdot \boxed{r_3} + 1 & \text{(remainder = 1)}
 \end{array}$$

For the factors, just use a calculator

Now using this work, solve for ever remainder, so in this case solve for

$$r_1, r_2, r_3, 1$$

Using the lines where they're the remainder

In order, we get

$$\begin{array}{ll}
 \boxed{r_1} = \boxed{m} - k_1 \cdot \boxed{a} & \text{(from first)} \\
 \boxed{r_2} = \boxed{a} - k_2 \cdot \boxed{r_1} & \text{(from second)} \\
 \boxed{r_3} = \boxed{r_1} - k_3 \cdot \boxed{r_2} & \text{(from third)} \\
 \boxed{1} = \boxed{r_2} - k_4 \cdot \boxed{r_3} & \text{(from last)}
 \end{array}$$

Now just work backwards and substitute each **divisor**, then combining like terms, but DO NOT actually multiply

Do this until you get to the first line with $\boxed{m}$. The inverse is the other factor with $\boxed{a}$

Example 41 (Euclidean Algorithm Practice)

Let's do some examples of the Euclidean Algorithm

Let's find the multiplicative inverse of $\boxed{37} \bmod \boxed{200}$

$$\begin{aligned}
 200 &= 5 \cdot 37 + 15 \\
 37 &= 2 \cdot 15 + 7 \\
 15 &= 2 \cdot 7 + 1
 \end{aligned}$$

Now that we found remainder = 1, let's solve for each remainder

$$\begin{aligned}
 15 &= 200 - 5 \cdot 37 \\
 7 &= 37 - 2 \cdot 15 \\
 1 &= 15 - 2 \cdot 7
 \end{aligned}$$

Now let's work backwards and substitute, starting with the bottom equation

$$\begin{aligned}
 1 &= 15 - 2(37 - 2 \cdot 15) \\
 1 &= 15 - 2 \cdot 37 + 4 \cdot 15 \\
 1 &= 5 \cdot 15 - 2 \cdot 37 \\
 1 &= 5(200 - 5 \cdot 37) - 2 \cdot 37 \\
 1 &= 5 \cdot 200 - 25 \cdot 37 - 2 \cdot 37 \\
 1 &= 5 \cdot 200 - 27 \cdot 37
 \end{aligned}$$

From this, we see that

$$37^{-1} = -27$$

But we don't like negatives, so

$$\begin{array}{rcl}
 37^{-1} & \equiv & -27 \pmod{200} \\
 +0 & \equiv & 200 \pmod{200} \\
 \hline
 37^{-1} & \equiv & 173 \pmod{200}
 \end{array}$$

4.3.2 Greatest Common Divisor (GCD)

Let's focus our attention to a the **gcd**

Definition 40 (Greatest Common Divisor, gcd)

The Greatest Common Divisor, or gcd, gets the greatest common divisor, or factor, between two numbers

$$a, b \in \mathbb{Z}, \text{ denoted as } \gcd(a, b)$$

Such that $\gcd(a, b)$ satisfies

1. $\gcd(a, b) \mid a$ and $\gcd(a, b) \mid b$
2. $\gcd(a, b) = \max \{d \in \mathbb{Z}^+ \mid d \mid a \text{ and } d \mid b\}$

We can find the gcd by using the Euclidean Algorithm. Basically the GCD is the **last nonzero remainder**

So just do Euclidean Algorithm until the remainder is 0, don't stop at 1

Example 42 (GCD with Euclidean Algorithm)

Determine

$$\gcd(142856, 882)$$

$$\begin{aligned} 142856 &= 161 \cdot 882 + 854 \\ 882 &= 1 \cdot 854 + 28 \\ 854 &= 30 \cdot 28 + 14 \\ 28 &= 2 \cdot 14 + 0 \end{aligned}$$

Since we found remainder = 0, the GCD is the last nonzero remainder, which is **14** in this case

$$\gcd(142856, 882) = 14$$

Theorem 3 (Bézout's Theorem)

Basically

$$a, b \in \mathbb{Z}^+ \implies \exists x, y \in \mathbb{Z}, ax + by = \gcd(a, b)$$

Theorem 4

If $a \mid bc$ and $\gcd(a, c) = 1$ then $a \mid b$

Proof

Since $\gcd(a, c) = 1$ then by Bézout's

$$ax + cy = \gcd(a, c) = 1$$

Multiply both sides by b and we get

$$bax + bcy = b$$

Since $a \mid bc$, that means $a \mid bcy$, and $a \mid bax$, hence $a \mid (bax + bcy)$

$$\therefore a \mid b$$

Theorem 5

If $a, b, c \in \mathbb{Z}, m \in \mathbb{Z}^+$, with $\gcd(c, m) = 1$

Then $ac \equiv bc \pmod{m}$ implies $a \equiv b \pmod{m}$

Proof

$ac \equiv bc \pmod{m}$ implies $m \mid (ac - bc)$ or $m \mid (a - b)c$

By the previous theorem

$$m \mid (a - b)$$

Therefore

$$a \equiv b \pmod{m}, \text{ as desired}$$

Note

Division by an integer $c \pmod{m}$ works if and only if $\gcd(c, m) = 1$. You may need to use Euclidean Algorithm to find c^{-1}

4.3.3 Divisors

Let's look at how to find the number of divisors in a number n

Definition 41 (Prime Factorization)

Prime factorization is breaking up a number into prime factors. An example is

$$\begin{aligned}
 24 &= 6 \cdot 4 \\
 &= (3 \cdot 2)(2 \cdot 2) \\
 &= 3^1 \cdot 2^3
 \end{aligned}$$

The prime factors are 3 and 2.

In order to find the number of divisors, we must first break up n into its prime factors

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_k^{e_k}$$

p_k : Prime factor
 e_k : Occurences

The number of divisors is then

$$(e_1 + 1)(e_2 + 1) \dots (e_k + 1)$$

4.4 Section 4.4 - Linear Congruences

To think of linear congruences, let's start with an example

Example 43

A group of soldiers line up in rows of 8. However, there are 4 soldiers left over. When the soldiers line up in rows of 7, there are 5 soldiers left over. What is the least possible number of soldiers in the group?

Let x = number of soldiers

From the description, we know:

$$x \equiv 4 \pmod{8}$$

$$x \equiv 5 \pmod{7}$$

This also means that

$$\exists k_1 \in \mathbb{Z} \text{ such that } x = 8k_1 + 4$$

$$8k_1 + 4 \equiv 5 \pmod{7} \quad (\text{substitute } x)$$

$$8k_1 \equiv 1 \pmod{7}$$

$$k_1 \equiv 1 \pmod{7} \quad (\text{since } 8^{-1} \text{ is } 1)$$

Expanding that last equation, we get

$$\exists k_2 \in \mathbb{Z}, k_1 = 7k_2 + 1$$

Substituting k_1 back into the equation for x , we get

$$x = 8k_1 + 4 \quad (\text{original})$$

$$x = 8(7k_2 + 1) + 4 \quad (\text{substitute})$$

$$x = 56k_2 + 12$$

Since we're looking for the smallest number of soldiers, we can set $k_2 = 0$ and we get $x = 12$ as our final answer

This is actually an example of the **Chinese Remainder Theorem**

Definition 42 (Relatively Prime)

Two numbers $a, b \in \mathbb{Z}$ are **relatively prime** if

$$\gcd(a, b) = 1$$

Theorem 6 (Chinese Remainder Theorem)

Basically

Let $m_1, m_2, \dots, m_n \in \mathbb{Z}^+$ which are pairwise relatively prime (they're all relatively prime to each other) and $a_1, a_2, \dots, a_n \in \mathbb{Z}$, then the system

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$\vdots$$

$$x \equiv a_n \pmod{m_n}$$

Has a unique solution that's mod $M = m_1 \cdot m_2 \cdot \dots \cdot m_2$

Example 44

Let's start with the system

$$x \equiv 14 \pmod{16}$$

$$x \equiv 21 \pmod{31}$$

$$x \equiv 12 \pmod{41}$$

We start by rewriting the first equation

$$x = 16k_1 + 14$$

And substituting it into the second equation

$$16k_1 + 14 \equiv 21 \pmod{31}$$

Now isolate k_1 . You need to find 16^{-1} , but thankfully we have some shortcuts

$$16k_1 \equiv 7 \pmod{31}$$

$$32k_1 \equiv 14 \pmod{31}$$

$$k_1 \equiv 14 \pmod{31}$$

(multiply 2)

$$(32 \equiv 1 \pmod{31})$$

Now that we have k_1 , let's expand it

$$k_1 = 31k_2 + 14$$

And plug it back into our equation of x

$$x = 16k_1 + 14$$

(original)

$$x = 16(31k_2 + 14) + 14$$

$$x = 16 \cdot 31k_2 + 16 \cdot 14 + 14$$

$$x = 496k_2 + 238$$

Now that we have a new equation for x involving k_2 , let's use our last equation

$$\begin{aligned}
 x &\equiv 12 \pmod{41} && \text{(last equation)} \\
 496k_2 + 238 &\equiv 12 \pmod{41} \\
 496k_2 &\equiv -226 \pmod{41} \\
 4k_2 &\equiv 20 \pmod{41} && \text{(reduce 496 and -226)}
 \end{aligned}$$

Now we gotta find $4^{-1} \pmod{41}$

Let's use the Euclidean Algorithm

$$\begin{aligned}
 41 &= 10 \cdot 4 + 1 \\
 1 &= 41 - 10 \cdot 4 \\
 4^{-1} &\equiv -10 \pmod{41} \\
 \boxed{4^{-1} &\equiv 31 \pmod{41}} && \text{(add 41)}
 \end{aligned}$$

Now that we have 4^{-1} , we solve

$$\begin{aligned}
 4k_2 &\equiv 20 \pmod{41} \\
 4^{-1} \cdot 4k_2 &\equiv 4^{-1} \cdot 20 \pmod{41} \\
 31 \cdot 4k_2 &\equiv 31 \cdot 20 \pmod{41} \\
 k_2 &\equiv 620 \pmod{41} \\
 \boxed{k_2 &\equiv 5 \pmod{41}}
 \end{aligned}$$

Now that we solved for k_2 , let's expand it one last time

$$k_2 = 41k_3 + 5$$

And plug it back into our equation for x involving k_2

$$x = 496k_2 + 238 \quad \text{(original)}$$

$$x = 496(41k_3 + 5) + 238 \quad \text{(substituted)}$$

$$\boxed{x = 20336k_3 + 2718} \quad \text{(simply)}$$

Now that we simplified for the equation, we see that it represents

$$\boxed{x \equiv 2718 \pmod{20336}}$$

And you can see from the Chinese Remainder Theorem that our mod is actually the product of our moduli

$$M = 20336 = 16 \cdot 31 \cdot 41$$

That was... very long, so let's write out the steps:

4.4.1 Steps for Solving Linear Congruences

Given the system: (I'm just gonna use up to 3 for this)

$$x \equiv a_1 \pmod{m_1}$$

$$x \equiv a_2 \pmod{m_2}$$

$$x \equiv a_3 \pmod{m_3}$$

The system has a **unique solution** if m_1, m_2, m_3 are pairwise coprime. If not, then it *could possibly* have a solution but not necessarily

1. Expand the first equation

$$x \equiv a_1 \pmod{m_1}$$

implies

$$\boxed{x = m_1 k_1 + a_1}, k_1 \in \mathbb{Z}$$

2. Plug that into the second equation

$$\begin{aligned}x &\equiv a_2 \pmod{m_2} \\ m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2}\end{aligned}$$

And solve for k_1

$$\begin{aligned}m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \\ m_1 k_1 &\equiv a_2 - a_1 \pmod{m_2} \\ m_1^{-1} k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\ k_1 &\equiv m_1^{-1}(a_2 - a_1) \pmod{m_2} \\ \boxed{k_1 \equiv c_1} &\pmod{m_2} \quad (c_1 \text{ is shorthand})\end{aligned}$$

Note

This isn't a formula, just steps with arbitrary variables. You'll need to find the inverse of m_1

And expand that last equation

$$k_1 \equiv c_1 \pmod{m_2}$$

implies

$$\boxed{k_1 = m_2 k_2 + c_1}, \quad k_2 \in \mathbb{Z}$$

3. Plug that into our equation for x

$$\begin{aligned}x &= m_1 k_1 + a_1 \\ \boxed{x &= m_1(m_2 k_2 + c_1) + a_1}\end{aligned}$$

Plug that into the last equation

$$m_1(m_2k_2 + c_1) + a_1 \equiv a_3 \pmod{m_3}$$

and solve for $k_2 \pmod{m_3}$

$$\vdots$$

$$\boxed{k_2 \equiv c_2 \pmod{m_3}}$$

Expand that equation

$$\boxed{k_2 = m_3k_3 + c_2}, k_3 \in \mathbb{Z}$$

4. Finally, plug that last equation into our SECOND equation for x

$$x = m_1(m_2k_2 + c_1) + a_1$$

$$x = m_1(m_2(m_3k_3 + c_2) + c_1) + a_1$$

$$\boxed{x = m_1m_2m_3k_3 + c_3}$$

Now we UN-expand this equation and see that

$$\boxed{x \equiv c_3} \pmod{m_1m_2m_3}$$

Basically, starting with the first equation

1. Expand first, solving for x
2. Plug x into second
3. Solve for k_1
4. Expand k_1
5. Plug k_1 back into first expanded of x (one with k_1)

6. Solve for k_2
7. Expand k_2
8. Plug k_2 back into second expanded of x (one with k_2)
9. Unexpand back into modulus

Just do these lines for the first equation, and these lines for each other equation

Let's do something else now

4.4.2 Factorials

1. $1! \pmod{2} = 1$
2. $2! \pmod{3} = 2$
3. $3! \pmod{4} = 2$
4. $4! \pmod{5} = 4$
5. $5! \pmod{6} = 0$, $5! = 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{6} = 0$
6. $6! \pmod{7} = 6$
7. $7! \pmod{8} = 0$, $7! = 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{8} = 0$
8. $8! \pmod{9} = 0$, $8! = 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{9} = 0$
9. $9! \pmod{10} = 0$, $9! = 9 \cdot 8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1 \pmod{10} = 0$
10. $10! \pmod{11} = 10$
11. $999! \pmod{1000} = 0$ since 1000 is composite

Notice that for all the ones that are not 0, the modulus is prime (except 4)

This brings us to our next theorems

4.4.3 Wilson's Theorem

Theorem 7 (Wilson's Theorem)

If p is a prime integer, then

$$(p-1)! \equiv p-1 \pmod{p}$$

or

$$(p-1)! \equiv -1 \pmod{p}$$

But for composite numbers, we have

Theorem 8 (Some Unnamed Theorem)

If n is a composite integer greater than 4,
then

$$(n-1)! \equiv 0 \pmod{n}$$

If $n = 4$, then

$$3! \equiv 2 \pmod{4}$$

4.4.4 Fermat's Little Theorem

There's examples that derive this, but I'm too lazy

Theorem 9 (Fermat's Little Theorem)

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

Proof

Consider $S = \{1, 2, \dots, p-1\}, p \in \mathbb{Z}$

Let's define a function $f : S \rightarrow S, f(x) = ax \pmod{p}$

Example 45

Let $p = 7, a = 3$

Then $S = \{1, 2, 3, 4, 5, 6\}$ and $f(x) = 3x \pmod{7}$

x	$f(x)$
1	3
2	6
3	2
4	5
5	1
6	4

We see that f is a **bijection**

so we claim f is a **bijection**

Why?

Let $y \in S$

We must solve

$$ax \equiv y \pmod{p}$$

and since $\gcd(a, p) = 1$, then a^{-1} exists $\pmod{p}$

Therefore, $a^{-1}(ax) \equiv a^{-1}y \pmod{p}$

or

$$x \equiv a^{-1}y \pmod{p}$$

we conclude that f is **onto (surjective)**

For the injective part, suppose $f(x_1) = f(x_2)$

Then

$$ax_1 \equiv ax_2 \pmod{p}$$

Since

$$\gcd(a, p) = 1, \text{ then } a^{-1} \text{ exists } \pmod{p}$$

Therefore

$$a^{-1}(ax_1) \equiv a^{-1}(ax_2) \pmod{p}$$

or

$$x_1 \equiv x_2 \pmod{p}$$

We conclude f is **one to one (injective)**

Now let's consider the product $f(1) \cdot f(2) \cdot \dots \cdot f(p-1)$

$$f(1) \cdot f(2) \cdot \dots \cdot f(p-1) \equiv 1 \cdot 2 \cdot 3 \cdot 4 \cdot 5 \cdot 6 \pmod{p}$$

$$a \cdot 2a \cdot \dots \cdot (p-1)a \equiv 1 \cdot 2 \cdot \dots \cdot (p-1) \pmod{p}$$

$$a^{p-1} [1 \cdot 2 \cdot \dots \cdot (p-1)] \equiv 1 \cdot 2 \cdot \dots \cdot (p-1) \pmod{p}$$

By Wilson's Theorem we get

$$(p-1)! \equiv p-1 \pmod{p}$$

$$\begin{aligned}
a^{p-1}! &\equiv (p-1)! \pmod{p} \\
a^{p-1}(p-1) &\equiv p-1 \pmod{p} \\
a^{p-1} &\equiv 1 \pmod{p} \text{ as desired}
\end{aligned}$$

Let's do some examples

Example 46

Compute the following

$$\begin{aligned}
&9^{2026} \pmod{11} \\
&a : 9 \\
&p : 11
\end{aligned}$$

$$\gcd(9, 11) = 1$$

By Fermat's Little Theorem,

$$9^{10} \equiv 1 \pmod{11}$$

We can use this to break up 2026

$$\begin{aligned}
9^{2026} &\equiv 9^{202 \cdot 10 + 6} \pmod{11} \\
&\equiv 9^{202 \cdot 10} \cdot 9^6 \pmod{11} \\
&\equiv 9^6 \pmod{11}
\end{aligned}$$

Remember that we can subtract 11 from 9, so $9 \equiv -2 \pmod{11}$

This makes

$$\begin{aligned}
&\equiv 9^6 \pmod{11} \\
&\equiv (-2)^6 \pmod{11} \\
&\equiv 64 \pmod{11} \\
&\equiv 9 \pmod{11}
\end{aligned}$$

4.4.5 Extended Fermat's Little Theorem

There's also an extension of this

Theorem 10 (Extended Fermat's Little Theorem)

If $a, p, q \in \mathbb{Z}$ with

1. p and q are distinct primes
2. $\gcd(a, pq) = 1$

Then

$$a^{(p-1)(q-1)} \equiv 1 \pmod{pq}$$

Example 47

Compute

1. $13^{2026} \pmod{253}$

First, let's check:

- (a) $253 = 11 \cdot 23$
 $p = 11, q = 23$, are distinct primes
- (b) $\gcd(13, 253) = 1$

This means we can use Extended Fermat's Little Theorem (EFLT)

$$\begin{aligned}
13^{(11-1)(23-1)} &\equiv 1 \pmod{253} \\
13^{10 \cdot 22} &\equiv 1 \pmod{253} \\
13^{220} &\equiv 1 \pmod{253}
\end{aligned}$$

Breaking up 2026,

$$2026 = 9 \cdot 220 + 46 \pmod{253}$$

This means

$$13^{2026} \equiv 13^{46} \pmod{253}$$

$$13 \pmod{253} = 13$$

$$\begin{aligned} 13^2 \pmod{253} &= 13^2 \\ &= 169 \end{aligned}$$

$$\begin{aligned} 13^4 \pmod{253} &= 169^2 \pmod{253} \\ &= 225 \end{aligned}$$

$$\begin{aligned} 13^8 \pmod{253} &= 225^2 \pmod{253} \\ &= 25 \end{aligned}$$

$$\begin{aligned} 13^{16} \pmod{253} &= 25^2 \pmod{253} \\ &= 119 \end{aligned}$$

$$\begin{aligned} 13^{32} \pmod{253} &= 119^2 \pmod{253} \\ &= 246 \end{aligned}$$

Now let's break up 46 into powers of 2

$$\begin{aligned} 13^{2026} &\equiv 13^{46} \\ &\equiv 13^{32} \cdot 13^8 \cdot 13^4 \cdot 13^2 \pmod{253} \\ &\equiv 246 \cdot 25 \cdot 225 \cdot 169 \pmod{253} \\ &\equiv 31 \end{aligned}$$

2.

$$24^{2026} \pmod{31}$$

(a) $p = 31$ is prime

(b) $\gcd(24, 31) = 1$

so FLT applies

$$24^{31-1} \equiv 1 \pmod{31}$$

$$24^{30} \equiv 1 \pmod{31}$$

$$2026 = 67 \cdot 30 + 16$$

$$\begin{aligned}
 24^{2026} &\equiv 24^{67 \cdot 30 + 16} \pmod{31} \\
 &\equiv (24^{30})^{67} \cdot 24^{16} \pmod{31} \\
 &\equiv 1^{67} \cdot 24^{16} \pmod{31} \\
 &\equiv 24^{16} \pmod{31} \\
 &\equiv 7 \pmod{31} \qquad \text{(brute forced)}
 \end{aligned}$$

4.4.6 An Implication of Fermat's Little Theorem

Definition 43 (Implication of FLT)

I'm not sure if this is called something but recall that Fermat's Little theorem states that

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

But actually, any multiple of $p - 1$ will make a equivalent to 1 mod p

We can rewrite this as

$$a^{k(p-1)} \equiv 1 \pmod{p}, k \in \mathbb{Z}^+$$

But this another way of saying this is

$$a^k \equiv 1 \pmod{p}, k \equiv 0 \pmod{p-1}$$

The second half just means that k is a multiple of $p - 1$

So TDLR, if a is raised to a multiple of $p - 1$, then it is equivalent to 1 mod p

This also implies that

$$a^x \equiv a^y \pmod{p} \text{ if } x \equiv y \pmod{p-1}$$

We'll see this being used later

4.6 Section 4.6 - Cryptography

Sorry in advance to anyone reading this since the professor never really gave a proper intro to this section and I'm too lazy right now to write an actual intro

But basically this section is about methods of encrypting data

Let's start with defining our messages

<u>A</u>	<u>B</u>	<u>C</u>	<u>D</u>	<u>E</u>
00	01	02	03	04
F	G	H	I	J
<u>05</u>	<u>06</u>	<u>07</u>	<u>08</u>	<u>09</u>
K	L	M	N	O
<u>10</u>	<u>11</u>	<u>12</u>	<u>13</u>	<u>14</u>
P	Q	R	S	T
<u>15</u>	<u>16</u>	<u>17</u>	<u>18</u>	<u>19</u>
U	V	W	X	Y
<u>20</u>	<u>21</u>	<u>22</u>	<u>23</u>	<u>24</u>
				Z
				25

Definition 44 (Encryption)

Given a message P (plain text), like **H E L L O**

using some method b to change the message into C (cyphertext)

Example 48

Let's use P from earlier, **H E L L O**, and get C by shifting the number value of each letter by +1

<u>H</u>	<u>E</u>	<u>L</u>	<u>L</u>	<u>O</u>
07	04	11	11	14
		↓ ₊₁		
<u>I</u>	<u>F</u>	<u>M</u>	<u>M</u>	<u>P</u>
08	05	12	12	15

or you could do

$$x \mapsto ax + b$$

This is called the **Caesar Cipher**

Definition 45 (Exponentiation Cycle)

Let's define this by doing an example

Example 49

Let's call P M for this example (M for message)

We need to choose a prime number bigger than 26 (since there are 26 letters)

Let's do $p = 43$

and choose an exponent

Let's do $e = 5$

Remember that our encrypted message is C

Let's just do the $\underbrace{\text{H}}_{07}$ in HELLO since I'm lazy

Let's start by **encrypting** our message.

This means that

$$p = 43$$

$$e = 5$$

$$M = 7$$

$$\begin{aligned} C &\equiv M^e \pmod{p} \\ C &\equiv 7^5 \pmod{43} \\ &\equiv 16,807 \pmod{43} \\ &\equiv 37 \pmod{43} \end{aligned}$$

Boom, so 37 is our encrypted message that the receiver has to decrypt.

Now let's look at how to **decrypt** the message

We KNOW

$$C \equiv M^e \pmod{p}$$

But we WANT

$$C^d \equiv M \pmod{p}$$

(trying to find d) because we need to find M

Plugging the KNOW (first equation) into the WANT (second equation), we get

$$\begin{aligned}
(M^e)^d &\equiv M \pmod{p} \\
M^{ed} &\equiv M \pmod{p} \\
M^{ed} \cdot M^{-1} &\equiv M \cdot M^{-1} \pmod{p} \\
M^{ed-1} &\equiv 1 \pmod{p}
\end{aligned}$$

Recall 2

From 4.4, we saw that

$$a^k \equiv 1 \pmod{p}, k \equiv 0 \pmod{p-1}$$

In this case, $k = ed - 1$, and we see that

$$\begin{aligned}
ed - 1 &\equiv 0 \pmod{p-1} \\
\boxed{ed &\equiv 1 \pmod{p-1}}
\end{aligned}$$

And since we know e and p , we can find d

$$\begin{aligned}
ed &\equiv 1 \pmod{p-1} \\
5d &\equiv 1 \pmod{42} \\
5d &\equiv 85 \pmod{42} && (\text{add } 2 \cdot 42) \\
5d &\equiv 5 \cdot 17 \pmod{42} \\
\boxed{d &\equiv 17 \pmod{42}}
\end{aligned}$$

Note

We used a shortcut way of finding the inverse but you'd typically use the Euclidean Algorithm

Recall 3

Also, note that we found what d is $\pmod{p-1}$ and not $\pmod{p}$. From 4.4 again, we can see

$$a^x \equiv a^y \pmod{p} \text{ if } x \equiv y \pmod{p-1}$$

There's a lot of algebra to prove that we can use this value of d even though it's a different mod, but I'm too lazy to prove it so just trust

Okay so now that we know that $d \equiv 17 \pmod{42}$

From the decrypter's standpoint, let's recap what know so far

$$p = 43$$

$$e = 5$$

$$d = 17$$

$$C = 37$$

We want to find M

Let's use the equation from earlier

$$C^d \equiv M \pmod{p}$$

So now we just solve

$$37^{17} \equiv M \pmod{43}$$

$$\begin{aligned} 37^{17} &\equiv (-6)^{17} \pmod{43} \\ &\equiv -(6)^{17} \pmod{43} \end{aligned}$$

$$6 \pmod{43} = 6$$

$$6^2 \pmod{43} = 36$$

$$\begin{aligned} 6^4 \pmod{43} &= 36^2 \\ &= 6 \end{aligned}$$

$$6^8 \pmod{43} = 36$$

$$6^{16} \pmod{43} = 6$$

$$\begin{aligned}
37^{17} &\equiv -(6)^{16} \cdot 6 \pmod{43} \\
&\equiv -6 \cdot 6 \pmod{43} \\
&\equiv -36 \pmod{43} \\
&\equiv 7 \pmod{43}
\end{aligned}$$

Which is what we expect!

Definition 46 (RSA Encryption)

An extension of what we just did is **RSA Encryption**, which uses **Extended Fermat's Little Theorem** instead of normal FLT

The way it works is you choose 2 distinct primes, p and q

$$p = 29, q = 47$$

$$\begin{aligned}
N &= pq = 29 \cdot 47 \\
&= 1363
\end{aligned}$$

And choose an exponent that is relatively prime of p and q

$$e = 5$$

Let's say our message is $\underbrace{H}_{07}$ again

$$M = 7$$

To encrypt, we do

$$C \equiv M^e \pmod{N}$$

Okay let's recap. We just went over 2 exponentiation cyphers

1. Given a prime p and exponent e that are relatively prime to each other (both public)

$$C \equiv M^e \pmod{p}$$

$$C^d \equiv M \pmod{p}$$

$$M^{ed} \equiv M \pmod{p}$$

$$M^{ed-1} \equiv 1 \pmod{p}$$

$$\implies ed \equiv 1 \pmod{p-1}$$

2. Given 2 distinct primes p, q and exponent e where $N = pq$ and e are both public

$$C \equiv M^e \pmod{N}$$

$$C^d \equiv M \pmod{N}$$

$$M^{ed} \equiv M \pmod{N}$$

$$M^{ed-1} \equiv 1 \pmod{N}$$

$$\implies ed \equiv 1 \pmod{(p-1)(q-1)}$$

Let's introduce another cypher, but let's start with an example

Example 50

Let's do HELLO again

$$\begin{array}{ccccc} \text{H} & \text{E} & \text{L} & \text{L} & \text{O} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 07 & 04 & 11 & 11 & 14 \end{array}$$

But let's put each letter into pairs (and let's add a random letter since HELLO is 5 letters long)

$\begin{array}{cc} \text{H} & \text{E} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 07 & 04 \end{array}$	$\begin{array}{cc} \text{L} & \text{L} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 11 & 11 \end{array}$	$\begin{array}{cc} \text{O} & \text{Z} \\ \underbrace{\hspace{1cm}} & \underbrace{\hspace{1cm}} \\ 14 & 25 \end{array}$
---	---	---

Let's now look at our first pair / block

H	E
07	04

$$\begin{aligned}
 M &= 0704 \\
 &= 704 \\
 p &= 43, q = 59 \\
 (p-1)(q-1) &= 42 \cdot 58 \\
 &= 2436 \\
 N &= pq = 2537 \\
 e &= 55
 \end{aligned}$$

Let's start by encrypting M , that is

$$\begin{aligned}
 C &\equiv M^e \pmod{N} \\
 C &\equiv 704^{55} \pmod{2537}
 \end{aligned}$$

Solving for C ...

Okay I'm too lazy to write it all out but just reduce 704^{55} into powers of 2 and you get

$$\begin{aligned}
 704^{55} &\equiv 637 \pmod{2537} \\
 C &\equiv \boxed{637} \pmod{2537}
 \end{aligned}$$

Now let's decrypt C , that is

$$M \equiv C^d \pmod{N}$$

To do this, we must solve

$$\begin{aligned}
 ed &\equiv 1 \pmod{(p-1)(q-1)} \\
 55d &\equiv 1 \pmod{2436}
 \end{aligned}$$

Now let's find 55^{-1}

We do this with the Euclidean Algorithm

$$2436 = 44 \cdot 55 + 16$$

$$55 = 3 \cdot 16 + 7$$

$$16 = 2 \cdot 7 + 2$$

$$7 = 3 \cdot 2 + 1$$

$$1 = 7 - 3 \cdot 2$$

$$2 = 16 - 2 \cdot 7$$

$$7 = 55 - 3 \cdot 16$$

$$16 = 2436 - 44 \cdot 55$$

$$1 = 7 - 3 \cdot (16 - 2 \cdot 7)$$

$$1 = 7 \cdot 7 - 3 \cdot 16$$

$$1 = 7(55 - 3 \cdot 16) - 3 \cdot 16$$

$$1 = 7 \cdot 55 - 24 \cdot 16$$

$$1 = 7 \cdot 55 - 24(2436 - 44 \cdot 55)$$

$$1 = 1063 \cdot 55 - 24 \cdot 2436$$

$$55^{-1} \equiv 1063 \pmod{2436}$$

$$d \equiv 1063 \pmod{2436}$$

$$M \equiv C^d \pmod{N}$$

$$M \equiv 637^{1063} \pmod{2537}$$

$$M \equiv 704 \pmod{2537}$$

Chapter 5

Section 5 - idk

5.1 Section 5.1 - Proof By Induction

This section really only covers one thing and that proving by induction

Let's start by proving that

$$\sum_{k=1}^n (2k-1) = n^2$$

Proof 1: Break everything up

$$\begin{aligned}\sum_{k=1}^n (2k-1) &= \sum_{k=1}^n (2k) - \sum_{k=1}^n 1 \\ &= 2 \sum_{k=1}^n k - \sum_{k=1}^n 1 \\ &= 2 \cdot \frac{n(n+1)}{2} - n \\ &= n^2 + n - n \\ &= n^2\end{aligned}$$

Proof 2: Represent as a picture (dots)

$$\sum_{k=1}^1 (2k-1) = 1 = 1^2$$

•

$$\sum_{k=1}^2 (2k-1) = 1 + 3 = 2^2$$

• •
• •

$$\sum_{k=1}^5 (2k-1) = 1 + 3 + 5 = 1^2$$

• • •
• • •
• • •

If $\sum_{k=1}^n (2k-1) = n^2$

$\overbrace{\begin{matrix} \bullet & \cdots & \bullet \\ \vdots & \ddots & \vdots \\ \bullet & \cdots & \bullet \end{matrix}}^n$

This is a square matrix so $n \times n$

Then

$$\sum_{k=1}^{n+1} (2k-1) = [1 + 3 + 5 + \dots + (2n-1)] + (2n+1)$$

$\overbrace{\begin{matrix} \bullet & \cdots & \bullet & \bullet \\ \vdots & \ddots & \vdots & \vdots \\ \bullet & \cdots & \bullet & \bullet \\ \bullet & \cdots & \bullet & \bullet \end{matrix}}^n$

The extra line is $n \times 1$ and $1 \times n$, so we have $2n$ more dots, and with the extra corner dot, we get

$$n^2 + 2n + 1 = (n+1)^2$$

But we're just assuming that the general formula $= n^2$ is true

Definition 47 (Proof by Induction)

Proof by Induction is basically proving that if a statement about the case n is true, then proving the case $n + 1$ will also prove every case

Let $P(n)$ be the proposition function

$$P(n) : \sum_{k=1}^n (2k-1) = n^2 \text{ for } n \in \mathbb{Z}^+$$

Base Case: $P(1)$ is true

Inductive Case: $P(n) \implies P(n+1)$

The above implies

$$P(1) \implies P(2)$$

$$P(2) \implies P(3)$$

$$\vdots$$

$$P(n) \implies P(n+1)$$

So we conclude that $P(n)$ is true $\forall n \in \mathbb{Z}^+$

This is specifically called **Weak Induction**

Strong Induction is

Base Case: $P(1)$ is true

Inductive Case: $P(1) \wedge P(2) \wedge \dots \wedge P(n) \implies P(n+1)$

Let's do some examples

Example 51

Prove the following

1.

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Proof

BASE CASE:

$$\begin{aligned}\sum_{k=1}^1 k^3 &= 1^3 = 1 \\ &= \left[\frac{n(n+1)}{2} \right]^2 = \left[\frac{1(1+1)}{2} \right]^2 = 1\end{aligned}$$

So the case $n = 1$ is true

INDUCTIVE CASE:

Assume that $\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$

We then prove $\sum_{k=1}^{n+1} k^3 = \left[\frac{(n+1)(n+2)}{2} \right]^2$

Let's do this by making the more complicated part (the proving part) equal to the simpler one

$$\begin{aligned}\sum_{k=1}^{n+1} k^3 &= \sum_{k=1}^n k^3 + (n+1)^3 = \sum_{k=1}^{n+1} k^3 = \left[\frac{n(n+1)}{2} \right]^2 + (n+1)^3 \\ &= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\ &= (n+1)^2 \cdot \left[\frac{n^2}{4} + (n+1) \right] \\ &= (n+1)^2 \cdot \left[\frac{n^2 + 4n + 4}{4} \right] \\ &= (n+1)^2 \cdot \frac{(n+2)^2}{4} \\ &= \frac{(n+1)^2(n+2)^2}{4} \\ &= \boxed{\left[\frac{(n+1)(n+2)}{2} \right]^2}\end{aligned}$$

2.

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6} \text{ for } n \in \mathbb{Z}^+$$

Proof

BASE CASE:

$$\sum_{k=1}^1 k^2 = 1^2 = 1$$

$$\frac{n(n+1)(2n+1)}{6} = \frac{1 \cdot (1+1)(2+1)}{6} = \frac{2 \cdot 3}{6} = 1$$

So out base case $n = 1$ is true

INDUCTIVE CASE:

Assume: $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$

We prove: $\sum_{k=1}^{n+1} k^2 = \frac{(n+1)(n+2)(2n+3)}{6}$

Again, lets start with the more complicated looking one and make it look like the simpler one

$$\begin{aligned} \sum_{k=1}^{n+1} k^2 &= \sum_{k=1}^n k^2 + (n+1)^2 \\ &= \frac{n(n+1)(2n+1)}{6} + \frac{6(n+1)^2}{6} \\ &= (n+1) \left[\frac{n(2n+1)}{6} + \frac{6(n+1)}{6} \right] \\ &= (n+1) \left[\frac{2n^2 + n + 6n + 1}{6} \right] \\ &= (n+1) \left[\frac{2n^2 + 7n + 1}{6} \right] \\ &= (n+1) \left[\frac{(n+2)(2n+3)}{6} \right] \\ &= \boxed{\frac{(n+1)(n+2)(2n+3)}{6}}, \text{ as desired} \end{aligned}$$

3.

$$2^n \leq n! \text{ for } n \geq 4$$

Proof

BASE CASE:

$$\begin{aligned}2^4 &= 16 \\4! &= 24 \\16 &< 24 \\\therefore 2^4 &\leq 4!\end{aligned}$$

So our base case $n = 4$ is true

INDUCTIVE CASE:

Assume: $2^n \leq n!$

We Prove: $2^{n+1} \leq (n+1)!$

$$\begin{aligned}2^{n+1} &= 2 \cdot 2^n \\2^{n+1} &\leq 2 \cdot n! \leq (n+1) \cdot n! && (\text{since } 2 \leq (n+1)) \\&= 2^{n+1} \leq (n+1)!\end{aligned}$$

Note

This one's kinda lame since we're dealing with inequalities

Let's do another proof but more words

Let's prove:

Every positive interger greater than 1 is a product of prime powers

Proof

BASE CASE: $n = 2$ is a prime power INDUCTIVE CASE:

Assume: n is a product of prime powers

We prove: $n + 1$ is also a product of prime powers

Case 1:

$n + 1$ is a prime

Then $n + 1$ is already a prime power

Case 2:

$n + 1$ is composite

Then $n + 1$ has two factors between 1 and n

1 and n are in our previous list which have prime powers

$\therefore n + 1$ also does

Example 52

Let's do an example with the Fibonacci sequence. Recall that

$$f_n : 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144, 233$$

Let's prove that every 4th Fibonacci number is divisible by 3

In other words,

$$3 \mid f_{4n}, n \in \mathbb{Z}^+$$

Proof

BASE CASE: $n = 1, f_4 = 3, 3 \mid 3$ so the case $n = 1$ is true

INDUCTIVE CASE:

Assume $3 \mid f_{4n}$

We prove $3 \mid f_{4(n+1)}$

$$\begin{aligned} f_{4(n+1)} &= f_{4n+4} = f_{4n+3} + f_{4n+2} \\ &= f_{4n+2} + f_{4n+1} + f_{4n+2} \\ &= 2f_{4n+2} + f_{4n+1} \\ &= 2(f_{4n+1} + f_{4n}) + f_{4n+1} \\ &= 3f_{4n+1} + 2f_{4n} \end{aligned}$$

Since $3 \mid 3f_{4n+1}$ and $3 \mid 2f_{4n}$, then $3 \mid 3f_{4n+1} + 2f_{4n}$

$\therefore 3 \mid f_{4(n+1)}$, as desired

Chapter 6

Section 6 - Counting

6.1 Section 6.1 - Intro to Counting

There are 2 highways from Palm Springs to Upland and 3 highways from Upland to Corona. How many ways can one travel from palm Springs to Corona if you must travel through Upland

	PU1	PU2
UC1		
UC2		
UC3		

$$2 \cdot 3 = 6 \text{ ways}$$

This is the fundamental theorem of Counting

Theorem 11 (Fundamental Theorem of Counting)

If there are a ways to do task A and b was of doing task B, then there are $a \cdot b$ ways to do task A and B

Example 53

Let's do some example of this

- How many 2 letter initials are there if...
 - The letters can be repeated?

$$\underbrace{26}_{\text{Letter 1}} \cdot \underbrace{26}_{\text{Letter 2}} = \boxed{676}$$

(b) The letters CANNOT be repeated?

$$\underbrace{26}_{\text{Letter 1}} \cdot \underbrace{25}_{\text{Letter 2}} = \boxed{650}$$

Since we can choose any letter for the first one, but the last one we can choose all but the one we just chose, so $26 - 1 = 25$

2. How many 7 digit phone numbers do not start with 0 or 1?

$$\underline{8} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} \cdot \underline{10} = \boxed{8,000,000}$$

3. How many ways can 7 people line up for a picture?

(a) There are no restrictions

$$\underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 7! = \boxed{5040}$$

(b) Dina must be the first person in the photo

$$\underline{1} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 6! = \boxed{720}$$

(c) Dina is NOT the first person in the photo

$$\underline{6} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 6 \cdot 6! = \boxed{4320}$$

OR

ways Dina w/ first + # ways Dina NOT first = total

ways Dina NOT first = total - # ways Dina w/ first

$$= 5040 - 720$$

$$= \boxed{4320}$$

(d) Dina must stand next to Omar

We do this by treating Dina and Omar as one pair

But since there's two ways of arranging Dina and Omar, Dina -> Omar and Omar -> Dina, we gotta multiply it every twice

$$2(\underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1}) = 2 \cdot 6! = \boxed{1440}$$

(e) Dina must not stand next to Omar

$$\begin{aligned}
 \# \text{ Dina next to Omar} + \# \text{ Dina NOT next to Omar} &= \text{total} \\
 \# \text{ Dina NOT next to Omar} &= \text{total} - \# \text{ Dina next to Omar} \\
 &= 5040 - 1440 \\
 &= \boxed{3600}
 \end{aligned}$$

4. A bit string is a sequence of 0's and 1's. How many bit strings are there of length 7?

(a) If there are no restrictions

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^7 = \boxed{128}$$

(b) Begin with 01

$$\underline{1} \cdot \underline{1} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^5 = \boxed{32}$$

We could also just say how many ways can we arrange a 5 length bit string since the first two are fixed (01)

(c) Are palindromes, ie are the same when written forwards or backwards

$$\underbrace{\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2}}_{\text{anything}} \cdot \underbrace{\underline{1} \cdot \underline{1} \cdot \underline{1}}_{\text{fixed}} = 2^4 = \boxed{16}$$

5. How many ways can 5 people order a single scoops of icecream if the possible flavors are vanilla, coffee, chocolate, strawberry, or olive?

$$\underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} = 5^5 = \boxed{3125}$$

Since there's 5 flavors and 5 people

6. How many whole numbers less than 100 are...

(a) Divisible by 3

$$\left\lfloor \frac{100}{3} \right\rfloor = 33 \# \text{ of possible ints div by 3}$$

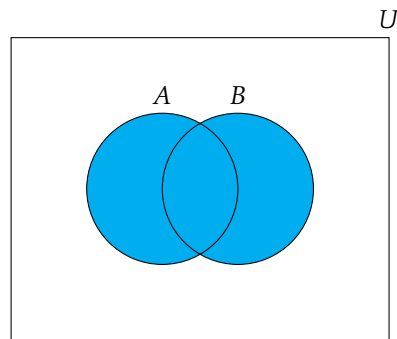
$$\left\lfloor \frac{100}{3} \right\rfloor + \underbrace{1}_{\text{including 0}} = \boxed{34}$$

(b) div by 7?

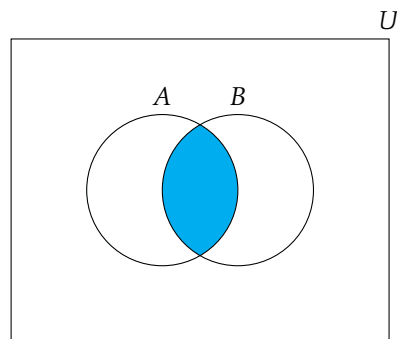
$$\left\lfloor \frac{100}{7} \right\rfloor + 1 = 14 + 1 = \boxed{15}$$

(c) div by 3 OR 7?

This represents 3 OR 7, where A is all numbers div by 3 and B is same but for 7



This represents div by BOTH 3 and 7



Recall 4

$$A \cup B = A + B - A \cap B$$

Being divisible by both 3 and 7 just means being divisible by 21

$$A \cap B = \left\lfloor \frac{100}{3 \cdot 7} \right\rfloor + 1 = 4 + 1 = 5$$

$$\begin{aligned} |A \cup B| &= |A| + |B| - |A \cap B| \\ &= 34 + 15 - 5 \\ &= \boxed{44} \end{aligned}$$

7. A die is rolled 5 times. How many ways can this happen if...

(a) No restrictions

$$\underline{6} \cdot \underline{6} \cdot \underline{6} \cdot \underline{6} \cdot \underline{6} = 6^5 = \boxed{7776}$$

(b) 4 is never rolled

$$\underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} \cdot \underline{5} = 5^5 = \boxed{3125}$$

(c) The number 4 is rolled at least once

ways 4 is rolled at least once + # never rolled = total

ways 4 is rolled at least once = total - # never rolled

$$= 7776 - 3125$$

$$= 6^5 - 5^5$$

$$= \boxed{4651}$$

8. A pair of dice is rolled. How many outcomes are possible if...

(a) No restrictions

$$\underline{6} \cdot \underline{6} = \boxed{36}$$

(b) The sum is 4

$$(1, 3), (3, 1), (2, 2)$$

so 3 ways

9. A coin is flipped 10 times. How many outcomes are possible if

(a) No restrictions

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^{10} = \boxed{1024}$$

(b) At least 1 of the flips is heads

$$\begin{aligned} \# \text{ ways no heads} + \# \text{ ways at least 1} &= \text{total} \\ \# \text{ at least 1} &= \text{total} - \underbrace{\# \text{ ways no heads}}_{\text{all tails}} \\ &= 2^{10} - 1 \\ &= \boxed{1023} \end{aligned}$$

6.1.1 Multiples of n up to k

Theorem 12 (Number of Multiples of n up to k)

To find the number of positive integers divisible by n up to k is given by

$$\left\lfloor \frac{k}{n} \right\rfloor$$

This DOES NOT include 0, so for all whole numbers it would be

$$\left\lfloor \frac{k}{n} \right\rfloor + 1$$

6.2 Section 6.2 - Pigeonhole Principle

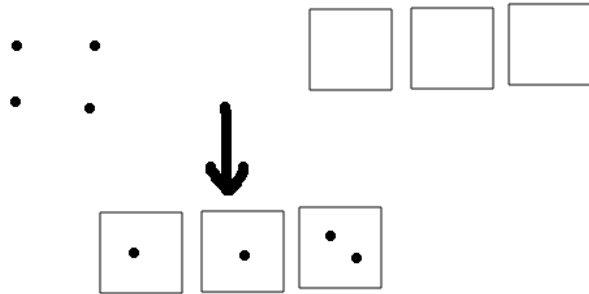
Definition 48 (Pigeonhole Principle)

If there are n boxes and more than n objects placed in these boxes, at least one box contains more than one object.

At least one box will contain

$$\left\lceil \frac{k}{n} \right\rceil$$

objects



Example 54

There are 43 people in this room. Are we guaranteed that

1. At least 2 people have the same birth month?

Solution:

Yes, since 12 months ; 43 people

6.3 Section 6.3 - Permutations and Combinations

Let's start with permutations

6.3.1 Permutations

Suppose you have 8 different books. How many ways can the books on a shelf be arranged if...

1. The shelf fits all 8 books?

$$\underline{8} \cdot \underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} \cdot \underline{3} \cdot \underline{2} \cdot \underline{1} = 8! = \boxed{40320}$$

2. The shelf fits 5 books

$$\underline{8} \cdot \underline{7} \cdot \underline{6} \cdot \underline{5} \cdot \underline{4} = \frac{8 \cdot 7 \cdot 6 \cdot 5 \cdot 4 \cdot 3 \cdot 2 \cdot 1}{3 \cdot 2 \cdot 1} = \frac{8!}{3!} = \frac{8!}{(8-5)!} = \boxed{6720}$$

Here, the order matters. We call this a **permutation**

Definition 49 (Permutation)

A **Permutation** is one specific way a group of items can be arranged where

order matters. In general, if we have n objects and we want to arrange k of them, then the number of ways is given by

$${}_nP_k = \frac{n!}{(n-k)!}$$

Example 55

How many ways can the president, vice president, and treasurer be assigned from a group of 4 people?

$$\underbrace{4}_P \cdot \underbrace{3}_{VP} \cdot \underbrace{2}_T = \boxed{24}$$

or, using the formula

$${}_4P_3 = \frac{4!}{(4-3)!} = \frac{4!}{1!} = \boxed{24}$$

Now let's move on to Combinations

6.3.2 Combinations

Example 56

How many committees of 3 people can be formed from 4 people?

Here, order **DOES NOT** matters

Here, there are 4 different committees

ABC	ABD	ACD	BCD
-----	-----	-----	-----

You can arrange the 3 people 6 different ways, given by

$${}_3P_3 = 3! = 6$$

Note

I would write everything out but I'm too lazy

Another way of getting that there are 4 committees is

$$\frac{\text{total \# ordered arrangements}}{\text{\# ways of permuting that arrangement}} = \frac{24}{6} = \frac{\frac{4!}{(4-3)!}}{3!} = \frac{4!}{3!(4-3)!}$$

This is the number of **Combinations**

Definition 50 (Combination)

A **Combinaton** is just an arrangement where order does not matter.

The number of combinations of k items taken from n items is

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

You would say "n choose k"

Think of this as "how many ways can you arrange k things in n slots"

Example 57 1. How many ways can 2 cards be dealt from a standard deck of cards?

52 cards, want 2 cards

$$\begin{aligned}\binom{52}{2} &= \frac{52!}{2!52-2!} \\ &= \frac{52!}{2!50!} \\ &= \frac{52 \cdot 51}{2} \\ &= \boxed{1326}\end{aligned}$$

2. A coin is flipped 10 times. How many ways can exactly 4 of the flips be heads?

$$\begin{aligned}\binom{10}{4} &= \frac{10!}{4!6!} \\ &= \boxed{210}\end{aligned}$$

notice how this also works for 6 flips to be tails!

3. Ten distinct points are labeled on a circle. How many quadrilaterals can be drawn using these points?

$$\binom{10}{4} = \frac{10!}{4!6!} = \boxed{210}$$

There's a LOT more examples but I'm too lazy to write them all out. Just go back to the notes and write them out later

Let's prove that $\binom{n+1}{k} = \binom{n}{k-1} + \binom{n}{k}$

Proof

$$\binom{n}{k-1} + \binom{n}{k} = \frac{n!}{(k-1)!(n-(k-1))!} + \frac{n!}{k!(n-k)!}$$

=

uhhh do later (it's on the very back of 6.3)

6.4 Section 6.4 - Binomial Identities

$$\begin{aligned} (a+b)^0 &= 1 \\ (a+b)^1 &= 1a + 1b \\ (a+b)^2 &= 1a^2 + 2a^1b^1 + 1b^2 \\ (a+b)^3 &= 1a^3 + 3a^2b^1 + 3a^1b^2 + 1b^3 \\ (a+b)^4 &= 1a^4 + 4a^3b^1 + 6a^2b^2 + 4a^1b^3 + 1b^4 \\ (a+b)^5 &= 1a^5 + 5a^4b^1 + 10a^3b^2 + 10a^2b^3 + 5a^1b^4 + 1b^5 \\ &\vdots \\ (a+b)^{10} &= 1a^{10} + \underbrace{10}_{\binom{10}{9}} a^9b^1 + \underbrace{45}_{\binom{10}{8}} a^8b^2 + \underbrace{120}_{\binom{10}{7}} a^7b^3 + \underbrace{210}_{\binom{10}{6}} a^6b^4 + \underbrace{252}_{\binom{10}{5}} a^5b^5 + \underbrace{210}_{\binom{10}{4}} a^4b^6 + 120a^3b^7 + 45a^2b^8 + \dots \end{aligned}$$

For all of these, it's 10 choose either the power of a or the power of b . This is because the power is essentially the number of arrangements of $aaabbbbbbb$ for example

This brings us to our next theorem

6.4.1 Binomial Theorem

Theorem 13 (Binomial Theorem)

$$\text{For } n = 0, 1, 2, \dots, (a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

I would put the proof here but it's long and I'm lazy

Example 58 1. A pizza parlor has 10 toppings. How many pizzas are possible if any number of toppings can be chosen but none can be repeated?

$$\begin{aligned} & \underbrace{\# 0 \text{ toppings}}_{\binom{10}{0}} + \underbrace{\# 1 \text{ topping}}_{\binom{10}{1}} + \dots + \underbrace{\# 10 \text{ toppings}}_{\binom{10}{10}} \\ &= \sum_{k=0}^{10} \binom{10}{k} \underbrace{1^{10-k} \cdot 1^k}_{\text{to use B.T}} \\ &= (1 + 1)^{10} \\ &= 2^{10} \\ &= \boxed{1024} \end{aligned}$$

2. How many bit strings are there of length 10?

$$\underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} \cdot \underline{2} = 2^{10} = \boxed{1024}$$

or

Place k 0's in the bit string.

$$\binom{10}{0} + \binom{10}{1} + \binom{10}{2} + \dots + \binom{10}{10} = (1 + 1)^{10} = 2^{10} = 1024$$

3. How many subsets does a set with 10 elements have?

uhhhh same thing

6.4.2 Sum of Combinations

Theorem 14 (Sum of Combinations)

$$\text{For } n = 0, 1, 2, \dots, \sum_{k=0}^n \binom{n}{k} = 2^n$$

Example 59

uhhhhh just use the formula

6.5 Section 6.5 - Stars and Bars

This section will just cover one idea, which is **Stars and Bars**

Let's start with an example

Example 60

Omar buys 9 icecreams. If the flavors available are chocolate and vanilla, how many possible orders can Omar make? An order in the sense like 5 chocolate 4 vanilla or 4 vanilla 5 chocolate

Let's brute force this first

Chocolate	Vanilla
0	9
1	8
2	7
3	6
4	5
5	4
6	3
7	2
8	1
9	1

So there are 10 orders possible.

Let's do this analytically.

We are essentially solving

$$a + b = 9$$

Where a and b are whole numbers (includes 0).

Thinking of this as symbols, we can write

$$\underbrace{\star\star\star\star | \star\star\star\star\star}_{10 \text{ symbols}}$$

Where the bar represents the divider between chocolate and vanilla and the stars represent the number of ice creams for each flavor.

We basically just have to count how many **arrangements** for the bar can be made. This is just a combination.

$$\binom{10}{1} = \frac{10!}{9! \cdot 1!} = \boxed{10}$$

Now that we have this example done, we can form the actual rule

Definition 51 (Stars and Bars)

If you have n slots with k options to choose from for each slot and

1. Order does not matter
2. All n slots must be filled
3. An option can be used 0 times

Then there are

$$\binom{n+k-1}{k-1} = \frac{(n+k-1)!}{n! \cdot (k-1)!}$$

Possibilities

Another way of thinking of this is stars and bars

Let

s : Stars, num slots
 b : Bars, options - 1

Then the combinations is

$$\binom{s+b}{b} = \frac{(s+b)!}{s! \cdot b!}$$

You are essentially solving for

$$x_1 + x_2 + \dots + x_k = n$$

Where

$$x_1, x_2, \dots, x_k \geq 0$$

In general, if the options must be used at least c times such that

$$x_1, x_2, \dots, x_k \geq c$$

Then the possibilities is just

$$\binom{n + k(1 - c) - 1}{k - 1}$$

For instance, if

$$x_1, x_2, \dots, x_k \geq 1$$

Possibilities are

$$\binom{n - 1}{k - 1}$$

Chapter 7

Section 7 - Discrete Probability

7.1 Section 7.1 - Intro to Probability

7.1.1 Event and Sample Space

Definition 52 (Event and Sample Space)

Let S be a finite, non-empty set called a **sample space** and a set $E \subseteq S$ called the **event space**. The probability of E occurring is defined as

$$P(E) = \frac{|E|}{|S|}$$

Example 61

Let's do some examples

1. Determine the probability of rolling the following numbers when 2 dice are rolled
 - (a) 12

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 12, $|E| = 1$

$$\begin{aligned} P(E) &= \frac{|E|}{|S|} \\ &= \boxed{\frac{1}{36}} \end{aligned}$$

(b) 2

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 2, $|E| = 1$

$$\begin{aligned} P(E) &= \frac{|E|}{|S|} \\ &= \boxed{\frac{1}{36}} \end{aligned}$$

(c) 3

S = possible rolls of 2 dice, $|S| = 36$

E = rolls that give a 3, $|E| = 2, (1 + 2), (2 + 1)$

$$\begin{aligned} P(E) &= \frac{|E|}{|S|} \\ &= \frac{2}{36} \\ &= \boxed{\frac{1}{18}} \end{aligned}$$

(d) 4

$$\begin{aligned}
 |S| &= 36 \\
 E &= \{(1, 3), (2, 2), (3, 1)\} \\
 |E| &= 3 \\
 P(E) &= \frac{3}{36} \\
 &= \boxed{\frac{1}{12}}
 \end{aligned}$$

This is just

Sum of dice	Probability
2	$\frac{1}{36}$
3	$\frac{2}{36}$
4	$\frac{3}{36}$
5	$\frac{4}{36}$
6	$\frac{5}{36}$
7	$\frac{6}{36}$
8	$\frac{5}{36}$
9	$\frac{4}{36}$
10	$\frac{3}{36}$
11	$\frac{2}{36}$
12	$\frac{1}{36}$

2. Determine the probability of rolling a 2 when a single die is rolled

$$\begin{aligned}
 S &= \{1, 2, 3, 4, 5, 6\} \\
 E &= \{2\} \\
 P(\text{rolling } 2) &= \boxed{\frac{1}{6}}
 \end{aligned}$$

3. Determine the probability of not rolling a 2 when a die is rolled

$$\begin{aligned}
 S &= \{1, 2, 3, 4, 5, 6\} \\
 E &= S - \{2\} \\
 &= \{1, 3, 4, 5, 6\} \\
 P(\text{not rolling } 2) &= \boxed{\frac{5}{6}}
 \end{aligned}$$

or

$$\begin{aligned}
 P(\text{not rolling } 2) + P(\text{rolling } 2) &= 1 \\
 P(\text{not rolling } 2) &= 1 - P(\text{rolling } 2) \\
 &= 1 - \frac{1}{6} \\
 &= \boxed{\frac{5}{6}}
 \end{aligned}$$

4. A card is chosen from a standard 52 deck. Determine the probability that the card is

(a) An ace

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= 4 \\
 P(\text{ace}) &= \frac{4}{52} \\
 &= \boxed{\frac{1}{13}}
 \end{aligned}$$

(b) An spade

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= 13 \\
 P(\text{spade}) &= \frac{13}{52} \\
 &= \boxed{\frac{1}{4}}
 \end{aligned}$$

(c) A spade or an ace

$$\begin{aligned}
 |S| &= 52 \\
 |E| &= |\text{Ace} \cup \text{Spade}| \\
 &= |\text{Ace}| + |\text{Spade}| - |\text{Ace} \cap \text{Spade}| \\
 &= 4 + 13 - 1 \\
 &= 16
 \end{aligned}$$

$$\begin{aligned}
 P(\text{ace or spade}) &= \frac{16}{52} \\
 &= \boxed{\frac{4}{13}}
 \end{aligned}$$

7.1.2 Probability with Multiple Subsets (OR)

From these examples we see that if $E_1, E_2 \subseteq S$, then

$$\begin{aligned}
 P(E_1 \cup E_2) &= \frac{|E_1 \cup E_2|}{|S|} \\
 &= \frac{|E_1| + |E_2| - |E_1 \cap E_2|}{|S|}
 \end{aligned}$$

7.2 Section 7.2 - Probability Theory

We didn't actually go over much

7.2.1 Event Independence

Definition 53 (Event Independence)

If H and Q are events, we say H and Q are **independent** if and only if

$$P(H \cap Q) = P(H) \cdot P(Q)$$

Example 62

A card is selected from a standard 52 deck. Determine the probability that

1. The card is a heart

$$\begin{aligned} P(\heartsuit) &= \frac{\# \text{ hearts}}{\# \text{ cards}} \\ &= \frac{13}{52} \\ &= \boxed{\frac{1}{4}} \end{aligned}$$

2. The card is a queen

$$\begin{aligned} P(\text{Queen}) &= \frac{\# \text{ queens}}{\# \text{ cards}} \\ &= \frac{4}{52} \\ &= \boxed{\frac{1}{13}} \end{aligned}$$

3. The card is a heart and a queen

$$\begin{aligned} P(\heartsuit \text{ and queen}) &= \frac{\# \text{ heart and queen}}{\# \text{ cards}} \\ &= \frac{1}{52} \end{aligned}$$

Notice that

$$P(\heartsuit \text{ and queen}) = P(\heartsuit) \cdot P(\text{queen})$$

This means that H and Q are **independent**

I would put the examples leading up to the derivation of this next thing but I'm too lazy

7.2.2 Binomial Probability

Definition 54 (Binomial Probability)

Given a limited event (ie 2 outcomes) with probability p , Then

$$P(k \text{ successes in } n \text{ trials}) = \binom{n}{k} p^k \cdot (1-p)^{n-k}$$

Example 63

Let's do some examples

A biased coin has probability of 0.51 for landing heads when flipped. Determine the probability of the coin landing on heads exactly 6 times when the coin is flipped 10 times

$$\begin{aligned} p &= 0.51 \\ k &= 6 && \text{(successes)} \\ n &= 10 \end{aligned}$$

$$\begin{aligned} P(6 \text{ successes in } 10 \text{ trials}) &= \binom{10}{6} (0.51)^6 \cdot (1-0.51)^{10-6} \\ &\approx \boxed{0.213} \end{aligned}$$

Example 64

Let's do another

Determine the probability of rolling a 12 **at least once** when a pair of dice is rolled 20 times.

Since we have "at least" let's do the opposite

$$P(\text{never rolling 12 out of 20}) = \binom{20}{0} \left(\frac{1}{36}\right)^0 \cdot \left(\frac{35}{36}\right)^{20} \\ \approx 0.569\dots$$

$$P(\text{rolling 12 at least once out of 20}) = 1 - P(\text{never rolling 12 out of 20}) \\ \approx 1 - 0.569 \\ \approx \boxed{0.431}$$

7.4 Section 7.4 - Expected Value

Example 65

Roulette is a game where a ball is rolled on a circular track. There are 38 numbers in the track, 18 red, 18 black, 2 green. If you correctly guess the number the ball lands on, you win.

1. What is the probability that you win?

$$\boxed{\frac{1}{38}}$$

2. You play 10 times. Find the probability of winning 6 out of the 10 times.

This is just binomial probability

$$\binom{10}{6} \left(\frac{1}{38}\right)^6 \left(\frac{37}{38}\right)^4 \approx \boxed{0.0000006}$$

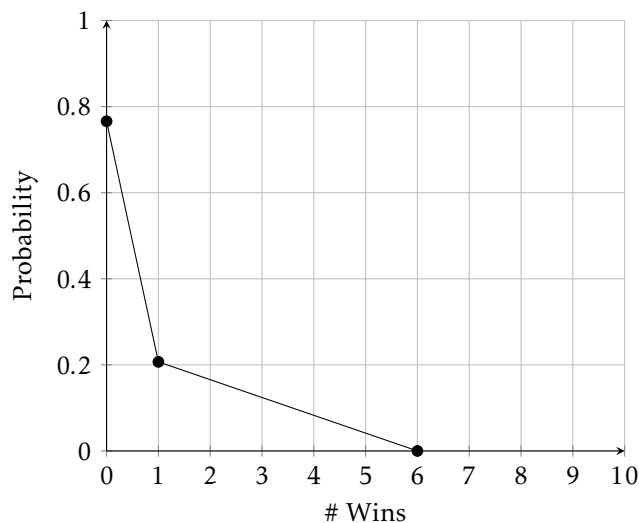
3. You play 10 times. Find the probability you never win

$$\left(\frac{37}{38}\right)^{10} \approx \boxed{0.766}$$

4. You play 10 times. Find the probability you win exactly once

$$\binom{10}{1} \left(\frac{1}{38}\right)^1 \left(\frac{37}{38}\right)^9 \approx \boxed{0.207}$$

Notice that if you graph the probability and wins, you get



5. You play 10 times. Find the probability of winning at least once
Let's use the compliment

$$\begin{aligned}
 P(\text{win at least once}) &= 1 - P(\text{never win in 10 spins}) \\
 &= 1 - \left(\frac{37}{38}\right)^{10} \\
 &\approx \boxed{0.234}
 \end{aligned}$$

In addition to guessing a number, you can also make a bet. The way it works is:

1. If you guess # correctly, then you win 35× your bet
2. If you guess incorrectly, then you lose your bet

Now we ask, how much do we win (or lose) **on average** when making a bet?

Assume we play a bunch of times. We bet \$1 every time

Let's play 38 times

Since the probability of winning is $\frac{1}{38}$, you'll probably win 1 time and lose 37 times

So

$$\frac{\text{total win/loss}}{\# \text{ played}} = \frac{\overbrace{1}^{\# \text{ Wins}} \cdot \overbrace{35}^{\$ \text{ gained}} + \overbrace{37}^{\# \text{ losses}} \cdot \overbrace{(-1)}^{\$ \text{ lost}}}{\underbrace{38}_{\text{played 38 times}}} = \boxed{-0.05}$$

Notice that we can simplify this into

$$\underbrace{\frac{1}{38}}_{P(\text{Win})} \cdot \overbrace{(35)}^{\$ \text{ Gained}} + \underbrace{\frac{37}{38}}_{P(\text{Loss})} \cdot \overbrace{(-1)}^{\$ \text{ Loss}}$$

This is called **expected value**

Definition 55 (Expected Value)

Expected Value is the what we expect to get from a probability outcome on average.

$$EV = \sum (\text{probability})(\text{payoff})$$

For every case

Example 66

Let's go back to the roulette example

We can also bet on red or black. If we choose correctly, we get paid the size of our bet. If we lose, we lose our bet.

What is the expected value of betting \$1 on red?

There are two cases.

1. We guess correctly, $P(\text{red}) = \frac{18}{38}$
2. We guess incorrectly, $P(\text{the rest}) = \frac{20}{38}$

$$\begin{aligned}
EV &= \sum (\text{probability})(\text{payoff}) \\
&= P(\text{red}) \cdot (1) + P(\text{the rest}) \cdot (-1) \\
&= \frac{20}{38} \cdot (1) + \frac{20}{38} \cdot (-1) \\
&= \frac{-2}{38} \\
&= \boxed{-0.05}
\end{aligned}$$

Let's just do one more

Example 67

Toys are randomly placed in cereal boxes so that 1 out of every 6 boxes contain a toy. How many boxes **on average** must be bought to get a toy?

There are an infinite amount of cases

1. First box contains toy, $P(1) = \frac{1}{6}$
2. Second box contains toy, $P(2) = \frac{5}{6} \cdot \frac{1}{6}$
3. Third box contains toy, $P(3) = \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6}$
4. You get the idea

We do $\frac{5}{6}$ because all the preceding boxes DO NOT contain toys, which has a $\frac{5}{6}$ chance.

Since the metric we are looking for is the number of boxes, the "payoff" is the number of boxes opened

$$\begin{aligned}
EV &= \sum (\text{probability})(\text{payoff}) \\
&= 1 \cdot \frac{1}{6} + 2 \cdot \frac{5}{6} \cdot \frac{1}{6} + 3 \cdot \left(\frac{5}{6}\right)^2 \cdot \frac{1}{6} + 4 \cdot \left(\frac{5}{6}\right)^3 \cdot \frac{1}{6} + \dots \\
&= \sum_{k=1}^{\infty} k \cdot \left(\frac{5}{6}\right)^{k-1} \cdot \frac{1}{6}
\end{aligned}$$

Recall 5

$$\sum_{k=0}^{\infty} k \cdot x^k = \frac{x}{(1-x)^2}, -1 < x < 1$$

I'm too lazy to write out the rest so

6

Chapter 8

Section 8 - Advanced Counting Techniques

8.1 Section 8.1 - Generating Functions

How do we determine an explicit formula for a recursive sequence?

For example,

$$\begin{aligned}f_0 &= 0 \\f_1 &= 1 \\f_{n+1} &= f_n + f_{n-1}, \quad n \geq 1\end{aligned}$$

Has an explicit function of

$$f_n = \frac{1}{\sqrt{5}} \left(\left(\frac{1 + \sqrt{5}}{2} \right)^n - \left(\frac{1 - \sqrt{5}}{2} \right)^n \right)$$

Let's consider a nicer example

$$\begin{aligned}a_0 &= 6 \\a_1 &= 13 \\a_n &= 5a_{n-1} - 6a_{n-2}, \quad n \geq 2\end{aligned}$$

Let's write out the first few terms

$$\begin{aligned} a_2 &= 5a_1 - 6a_0 \\ &= 5 \cdot 13 - 6 \cdot 5 \\ &= 29 \end{aligned}$$

$$\begin{aligned} a_3 &= 5 \cdot 29 - 6 \cdot 13 \\ &= 67 \end{aligned}$$

$$\begin{aligned} a_4 &= 5 \cdot 67 - 6 \cdot 29 \\ &= 161 \end{aligned}$$

We form a **generating function** for the sequence

$$a_0, a_1, a_2, \dots$$

Which is a power series whose coefficients are the sequence

$$f(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + \dots$$

From our example above, the generating function is

$$f(x) = 6 + 13x + 29x^2 + 67x^3 + 161x^4 + \dots$$

To reduce this, let's go back to our recursive function

$$a_n = 5a_{n-1} - 6a_{n-2}$$

Rearranging, we get

$$a_n - 5a_{n-1} + 6a_{n-2} = 0$$

Now let's bring in our generating function

$$\begin{aligned} f(x) &= 6 + 13x + \cancel{29x^2} + \cancel{67x^3} + \cancel{161x^4} + \dots \\ -5xf(x) &= -30x - \cancel{65x^2} - \cancel{145x^3} - \cancel{335x^4} + \dots \\ 6x^2f(x) &= \quad \quad \cancel{36x^2} + \cancel{78x^3} + \cancel{174x^4} + \dots \end{aligned}$$

$$\begin{aligned}
 (1 - 5x + 6x^2)f(x) &= 6 - 17x \\
 f(x) &= \frac{6 - 17x}{1 - 5x + 6x^2} \\
 &= \frac{6 - 17x}{(1 - 2x)(1 - 3x)} \\
 &= \frac{A}{1 - 2x} + \frac{B}{1 - 3x}
 \end{aligned}$$

$$A(1 - 3x) + B(1 - 2x) = 6 - 17x$$

$$\begin{aligned}
 \text{Let } x &= \frac{1}{3} \\
 B(1 - \frac{2}{3}) &= 6 - \frac{17}{3} \\
 B \cdot \frac{1}{3} &= \frac{18}{3} - \frac{17}{3} \\
 \boxed{B = 1}
 \end{aligned}$$

$$\begin{aligned}
 \text{Let } x &= \frac{1}{2} \\
 A(1 - \frac{3}{2}) &= 6 - \frac{17}{2} \\
 A(-\frac{1}{2}) &= \frac{12}{2} - \frac{17}{2} \\
 -\frac{1}{2}A &= -\frac{5}{2} \\
 \boxed{A = 5}
 \end{aligned}$$

So

$$\boxed{f(x) = \frac{5}{1 - 2x} + \frac{1}{1 - 3x}}$$

Recall 6

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots = \sum_{n=0}^{\infty} x^n$$

So

$$\begin{aligned} f(x) &= 5 \sum_{n=0}^{\infty} (2x)^n + \sum_{n=0}^{\infty} (3x)^n \\ \sum_{n=0}^{\infty} a_n x^n &= \sum_{n=0}^{\infty} (5 \cdot 2^n + 3^n) x^n \\ a_n &= 5 \cdot 2^n + 3^n \end{aligned}$$

But wait. If

$$\begin{aligned} a_n &= 5a_{n-1} - 6a_{n-2} \\ a_n - 5a_{n-1} + 6a_{n-2} &= 0 \\ \implies x^2 - 5x + 6 &= 0 \\ (x-3)(x-2) &= 0 \\ x &= 2, 3 \end{aligned}$$

But

$$a_n = 5 \cdot 2^n + 1 \cdot 3^n$$

How do we get the 5 and 1 (the coefficients)?

Using this shortcut, we see can see that

$$a_n = A \cdot 2^n + B \cdot 3^n$$

Let's use $a_0 = 6$ and $a_1 = 13$

$$a_0 = 6 = A \cdot 2^0 + B \cdot 3^0$$

$$6 = A + B$$

$$A = 6 - B$$

$$a_1 = 13 = A \cdot 2 + B \cdot 3$$

$$13 = 2A + 3B$$

$$13 = 2(6 - B) + 3B$$

$$13 = 12 - 2B + 3B$$

$$\boxed{B = 1}$$

$$A = 6 - 1$$

$$\boxed{A = 5}$$

Chapter 9

Exam Reviews

9.3 Review for Exam 3

9.3.1 Wilson's Theorem

Theorem 15 (Wilson's Theorem)

If p is a prime integer, then

$$(p-1)! \equiv p-1 \pmod{p}$$

or

$$(p-1)! \equiv -1 \pmod{p}$$

But for composite numbers, we have

Theorem 16 (Some Unnamed Theorem)

If n is a composite integer greater than 4,

then

$$(n-1)! \equiv 0 \pmod{n}$$

If $n = 4$, then

$$3! \equiv 2 \pmod{4}$$

9.3.2 Fermat's Little Theorem

Theorem 17 (Fermat's Little Theorem)

If $a \in \mathbb{Z}$ with,

1. p is prime
2. $\gcd(a, p) = 1$

$$a^{p-1} \equiv 1 \pmod{p}$$

9.3.3 Extended Fermat's Little Theorem

Theorem 18 (Extended Fermat's Little Theorem)

If $a, p, q \in \mathbb{Z}$ with

1. p and q are distinct primes
2. $\gcd(a, pq) = 1$

Then

$$a^{(p-1)(q-1)} \equiv 1 \pmod{pq}$$

9.3.4 Bezout's Theorem

Theorem 19 (Bézout's Theorem)

Basically

$$a, b \in \mathbb{Z}^+ \implies \exists x, y \in \mathbb{Z}, ax + by = \gcd(a, b)$$

When given a problem that's like find x, y in $ax + by = \gcd(a, b)$ (where they give you a and b), just do Euclidean and stop when you get to the actual GCD then back solve

9.3.5 Is a number prime or composite?

Lowkey just brute force up until

$$\lceil \sqrt{n} \rceil$$

9.3.6 Prime Factorization

Let's look at how to find the number of divisors in a number n

Definition 56 (Prime Factorization)

Prime factorization is breaking up a number into prime factors. An example is

$$\begin{aligned} 24 &= 6 \cdot 4 \\ &= (3 \cdot 2)(2 \cdot 2) \\ &= 3^1 \cdot 2^3 \end{aligned}$$

The prime factors are 3 and 2.

In order to find the number of divisors, we must first break up n into its prime factors

$$n = p_1^{e_1} \cdot p_2^{e_2} \cdot \dots \cdot p_k^{e_k}$$

p_k : Prime factor
 e_k : Occurrences

The number of divisors is then

$$(e_1 + 1)(e_2 + 1) \dots (e_k + 1)$$

9.3.7 Chinese Remainder Theorem

Given the system: (I'm just gonna use up to 3 for this)

$$\begin{aligned} x &\equiv a_1 \pmod{m_1} \\ x &\equiv a_2 \pmod{m_2} \\ x &\equiv a_3 \pmod{m_3} \end{aligned}$$

The system has a **unique solution** if m_1, m_2, m_3 are pairwise coprime. If not, then it *could possibly* have a solution but not necessarily

1. Expand the first equation

$$x \equiv a_1 \pmod{m_1}$$

implies

$$\boxed{x = m_1 k_1 + a_1}, k_1 \in \mathbb{Z}$$

2. Plug that into the second equation

$$\begin{aligned} x &\equiv a_2 \pmod{m_2} \\ m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \end{aligned}$$

And solve for k_1

$$\begin{aligned} m_1 k_1 + a_1 &\equiv a_2 \pmod{m_2} \\ m_1 k_1 &\equiv a_2 - a_1 \pmod{m_2} \\ m_1^{-1} k_1 &\equiv m_1^{-1} (a_2 - a_1) \pmod{m_2} \\ k_1 &\equiv m_1^{-1} (a_2 - a_1) \pmod{m_2} \\ \boxed{k_1 \equiv c_1} &\pmod{m_2} \quad (c_1 \text{ is shorthand}) \end{aligned}$$

Note

This isn't a formula, just steps with arbitrary variables. You'll need to find the inverse of m_1

And expand that last equation

$$k_1 \equiv c_1 \pmod{m_2}$$

implies

$$\boxed{k_1 = m_2 k_2 + c_1}, k_2 \in \mathbb{Z}$$

3. Plug that into our equation for x

$$x = m_1 k_1 + a_1$$

$$\boxed{x = m_1(m_2 k_2 + c_1) + a_1}$$

Plug that into the last equation

$$m_1(m_2 k_2 + c_1) + a_1 \equiv a_3 \pmod{m_3}$$

and solve for $k_2 \pmod{m_3}$

$$\vdots$$

$$\boxed{k_2 \equiv c_2 \pmod{m_3}}$$

Expand that equation

$$\boxed{k_2 = m_3 k_3 + c_2}, k_3 \in \mathbb{Z}$$

4. Finally, plug that last equation into our SECOND equation for x

$$x = m_1(m_2 k_2 + c_1) + a_1$$

$$x = m_1(m_2(m_3 k_3 + c_2) + c_1) + a_1$$

$$\boxed{x = m_1 m_2 m_3 k_3 + c_3}$$

Now we UN-expand this equation and see that

$$\boxed{x \equiv c_3 \pmod{m_1 m_2 m_3}}$$

Basically, starting with the first equation

1. Expand first, solving for x
2. Plug x into second
3. Solve for k_1
4. Expand k_1
5. Plug k_1 back into first expanded of x (one with k_1)
6. Solve for k_2
7. Expand k_2
8. Plug k_2 back into second expanded of x (one with k_2)
9. Unexpand back into modulus

Just do these lines for the first equation, and these lines for each other equation

9.4 Exam 4 Review

9.4.1 Binary and Base n

Write 1577 in binary

$$\begin{aligned}
 1577 &= 2 \cdot 788 + 1 \\
 788 &= 2 \cdot 394 + 0 \\
 394 &= 2 \cdot 197 + 0 \\
 197 &= 2 \cdot 98 + 1 \\
 98 &= 2 \cdot 49 + 0 \\
 49 &= 2 \cdot 24 + 1 \\
 24 &= 2 \cdot 12 + 0 \\
 12 &= 2 \cdot 6 + 0 \\
 6 &= 2 \cdot 3 + 0 \\
 3 &= 2 \cdot 1 + 1 \\
 1 &= 2 \cdot 0 + 1
 \end{aligned}$$

Writing the bottom to top left to right...

$$\boxed{11000101001}$$

We could check this by just doing

$$1577 = 2^{10} + 2^9 + \dots$$

Let's try representing 1577 in base 8. Remember that each remainder we get is the coefficient to the corresponding power

$$1577 = 8 \cdot 197 + \mathbf{1}$$

$$197 = 8 \cdot 24 + \mathbf{5}$$

$$24 = 8 \cdot 3 + \mathbf{0}$$

$$3 = 8 \cdot 0 + \mathbf{3}$$

Rewriting this, we get

$$\underbrace{3}_{8^3} \underbrace{0}_{8^3} \underbrace{5}_{8^2} \underbrace{1}_{8^0}$$

So

$$1577 = 3 \cdot 8^3 + 5 \cdot 8^1 + 1 \cdot 8^0$$

Now let's do it with base 16

$$1577 = 16 \cdot 98 + \mathbf{9}$$

$$98 = 16 \cdot 6 + \mathbf{2}$$

$$6 = 16 \cdot 0 + \mathbf{6}$$

$$\boxed{1577 = 629_{16}}$$

$$1577 = 6 \cdot 16^2 + 2 \cdot 16^1 + 9 \cdot 16^0$$

For our last example, write the $AB0CD_{16}$ in base 10 (aka normal numbers)

Recall that

$$A = 10$$

$$B = 11$$

$$\vdots$$

$AB0CD$ is just 10 11 0 12 13

Since there's 5 "numbers" that means we start with $16^{5-1} = 16^4$

$$\begin{aligned} AB0CD_{16} &= 10 \cdot 16^4 + 11 \cdot 16^3 + 0 \cdot 16^2 + 12 \cdot 16^1 + 13 \cdot 16^0 \\ &= \boxed{700621} \end{aligned} \quad \text{(just plug into calc)}$$

9.4.2 RSA and Encryption

FLT Encrytion

For Encryption using FLT, basically

We have:

M : Message

C : Encrypted Message

e : Encryption Key

d : Decryption Key

p : Prime mod

To **encrypt** a message M to C

$$C \equiv M^e \pmod{p}$$

To **decrypt** C back into M , we must first find d

$$ed \equiv 1 \pmod{p-1}$$

Just find e^{-1} using Euclidean and plug it into

$$M \equiv C^d \pmod{p}$$

FLT / RSA Encryption

We have:

M : Message
 C : Encrypted Message
 e : Encryption Key
 d : Decryption Key
 p : Prime mod 1
 q : Prime mod 2
 N : Product of $p \cdot q$

To **encrypt** a message M to C

$$C \equiv M^e \pmod{N}$$

To **decrypt** C back into M , we must first find d

$$ed \equiv 1 \pmod{(p-1)(q-1)}$$

Just find e^{-1} using Euclidean and plug it into

$$M \equiv C^d \pmod{N}$$

9.4.3 Proof By Induction

$$\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$$

Proof

BASE CASE:

$$\begin{aligned}\sum_{k=1}^1 k^3 &= 1^3 = 1 \\ &= \left[\frac{n(n+1)}{2} \right]^2 = \left[\frac{1(1+1)}{2} \right]^2 = 1\end{aligned}$$

So the case $n = 1$ is true

INDUCTIVE CASE:

Assume that $\sum_{k=1}^n k^3 = \left[\frac{n(n+1)}{2} \right]^2$

We then prove $\sum_{k=1}^{n+1} k^3 = \left[\frac{(n+1)(n+2)}{2} \right]^2$

Let's do this by making the more complicated part (the proving part) equal to the simpler one

$$\begin{aligned}\sum_{k=1}^{n+1} k^3 &= \sum_{k=1}^n k^3 + (n+1)^3 = \sum_{k=1}^{n+1} k^3 = \left[\frac{n(n+1)}{2} \right]^2 + (n+1)^3 \\ &= \frac{n^2(n+1)^2}{4} + (n+1)^3 \\ &= (n+1)^2 \cdot \left[\frac{n^2}{4} + (n+1) \right] \\ &= (n+1)^2 \cdot \left[\frac{n^2 + 4n + 4}{4} \right] \\ &= (n+1)^2 \cdot \frac{(n+2)^2}{4} \\ &= \frac{(n+1)^2(n+2)^2}{4} \\ &= \left[\frac{(n+1)(n+2)}{2} \right]^2\end{aligned}$$

9.4.4 Multiples of n up to k

Theorem 20 (Number of Multiples of n up to k)

To find the number of positive integers divisible by n up to k is given by

$$\left\lfloor \frac{k}{n} \right\rfloor$$

This DOES NOT include 0, so for all whole numbers it would be

$$\left\lfloor \frac{k}{n} \right\rfloor + 1$$

9.4.5 Pigeonhole Principle

Definition 57 (Pigeonhole Principle)

If there are n boxes and more than n objects placed in these boxes, at least one box contains more than one object.

At least one box will contain

$$\left\lceil \frac{k}{n} \right\rceil$$

objects

9.4.6 Permutations and Combinations

Permutations

Definition 58 (Permutation)

A **Permutation** is one specific way a group of items can be arranged where order matters. In general, if we have n objects and we want to arrange k of them, then the number of ways is given by

$${}_nP_k = \frac{n!}{(n-k)!}$$

Combinations

Definition 59 (Combination)

A **Combination** is just an arrangement where order does not matter.

The number of combinations of k items taken from n items is

$$\binom{n}{k} = {}_nC_k = \frac{n!}{k!(n-k)!}$$

You would say "n choose k"

Think of this as "how many ways can you arrange k things in n slots"

9.4.7 Binomial Theorem

Theorem 21 (Binomial Theorem)

$$\text{For } n = 0, 1, 2, \dots, (a + b)^n = \sum_{k=0}^n \binom{n}{k} a^{n-k} b^k$$

9.4.8 Sum of Combinations

Theorem 22 (Sum of Combinations)

$$\text{For } n = 0, 1, 2, \dots, \sum_{k=0}^n \binom{n}{k} = 2^n$$

9.4.9 Arrangements of Letters

For a word with k letters, where each letter has n_i appearances, the numbers of arrangements is given by

$$\frac{k!}{n_1! \cdot n_2! \cdot \dots \cdot n_k!}$$

You basically just do

$$\binom{k}{n_1} \cdot \binom{k-n_1}{n_2} \cdot \binom{k-n_1-n_2}{n_3} \cdot \dots$$

For a word with all unique letters, it's just

$$k!$$

9.4.10 Stars and Bars

When you have n slots and k options for each slot, the number of possible combinations is

$$\frac{(n+k-1)!}{n! \cdot (k-1)!} = \binom{n+k-1}{k-1}$$

Another way of thinking of this is stars and bars

Let

s : Stars, num slots

b : Bars, options - 1

Then the combinations is

$$\frac{(s+b)!}{s! \cdot b!} = \binom{s+b}{b}$$

9.4.11 Binomial Probability

Given a limited event (ie 2 outcomes) with probability p ,

$$p(k \text{ successes in } n \text{ trials}) = \binom{n}{k} \cdot p^k \cdot (1-p)^{n-k}$$

9.4.12 Expected Value (EV)

$$EV = \sum (\text{probability})(\text{payoff})$$